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HARVARD GRADUATE SCHOOL OF DESIGN | MASTER IN URBAN PLANNING '28

COLUMBIA UNIVERSITY | B.A. ECONOMICS–MATHEMATICAL PROBABILITY '26

"A CITY IS THE PLACE OF AVAILABILITIES. IT IS THE PLACE WHERE A SMALL BOY, AS HE WALKS THROUGH IT, MAY SEE SOMETHING THAT WILL TELL HIM WHAT HE WANTS TO DO HIS WHOLE LIFE."

— LOUIS KAHN

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PART 1: DATA-DRIVEN URBAN SYSTEMS & SPATIAL ANALYTICS

Modern urban strategy requires an urgent shift away from static, platform-bound mapping toward dynamic, accessible spatial intelligence systems. To navigate the compounding challenges of 21st-century cities, we need better frameworks to predict, monitor, and model the systems-level impacts of our infrastructure interventions.

My core thesis is that **the integration of artificial intelligence and complexity science is the key to renewing and advancing urban planning and design for the modern era**. This philosophy applies equally to optimizing physical transit grids, simulating agent-based behavioral dynamics, and anticipating how decentralized technologies like blockchain will reshape real estate, asset valuation, and property structures in the future.

The work in this section focuses on data engineering and custom pipeline architecture—specifically **mining unstructured data sources to uncover unique urban insights and generate new spatial models**. They demonstrate a continuous intent to build the adaptive spatial tools public agencies and private markets require to make **responsive, systemic, and forward-looking structural decisions**. In the future, I hope to contribute to building planning guardrails, spatial knowledge graphs and bayesian networks for spatial AI.

1.1

Open-Data 3D City Modeling

Digital Surface and Terrain Model Raster-to-Vector Differencing & Open Data Contribution Pipeline

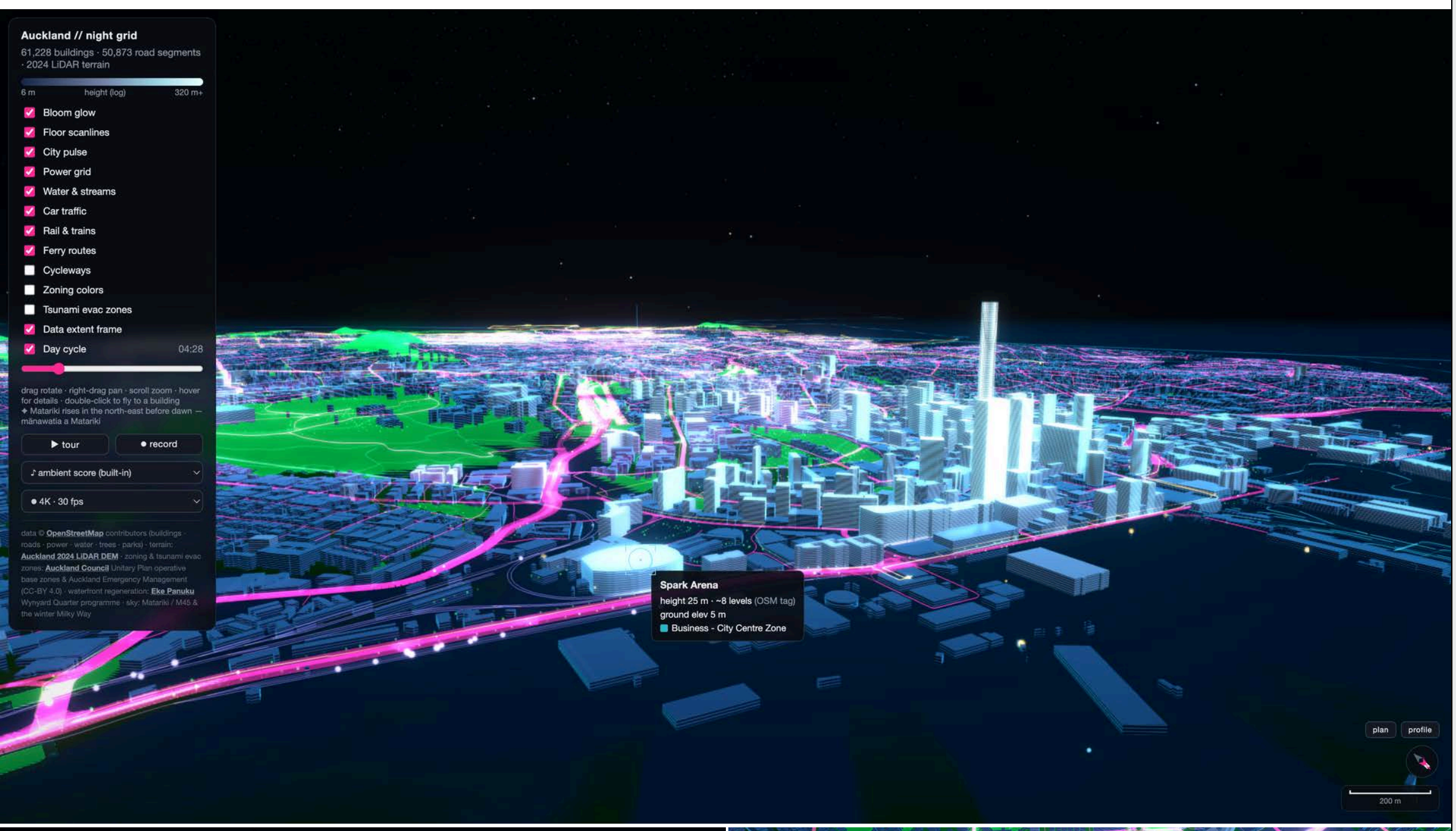
- **Metadata:** Independent Geospatial Product Design | June 2026 | Sole Author: Amy X. Chen | Live Interactive Web Application, Video, & GitHub Repository
- **Tech Stack:** Python 3.11, WebGL, Raster LiDAR Processing (DSM/DEM), GitHub Actions

Model: https://amyxqc.github.io/auckland-night-grid/
Video: https://youtu.be/feJJUionzyo
Documentation: github.com/amyxqc/auckland-night-grid

While 3D urban massing models are vital for simulating neighborhood density, microclimatic shadows, and real estate asset capacities, cities are often forced to choose between proprietary datasets that are heavy and costly to update, or flat 2D records. In Auckland, **OpenStreetMap (OSM) footprints** provide a comprehensive horizontal layout of the city, but their vertical data columns are mostly blank, leaving the city existing as flat outlines with no volume or terrain context.

To democratize 3D analytics, this project engineered a scalable Python data pipeline in the NZTM2000 coordinate system. The script handles spatial differences by overlaying 2D crowdsourced OSM shapes onto public municipal LiDAR cells, calculating the vertical delta between the Digital Surface Model (DSM) and Digital Terrain Model (DEM) to **derive precise above-ground heights across 61,228 individual building footprints**.

This proof-of-concept renders an entire metropolitan core fluidly within a basic web browser interface, eliminating the need for expensive proprietary software licenses. Because it utilizes standard municipal releases, the methodology provides a transferable, lightweight framework that any public agency can use to integrate global open data with municipal data sets.







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1.3

A TIME SERIES ESTIMATE OF PROTECTED LANE RETURNS ON CITI BIKE DEMAND

The NYC Streets Plan requires the City of New York to install at least 50 miles of protected bike lanes in each year from 2023 to 2026 and a total of 250 miles in the five years from 2022 to 2026.

To **estimate the effect of bike lane expansion on actual ridership**, I cleaned and built a weekly time series dataset (2022–2024) from Citi Bike trip records using Python and GitHub, integrating station-level activity, weather data from the National Climatic Data Center, and protected bike lane installation data from Transportation Alternatives' Protected Bike Lane Tracker. **Applied regression models with autoregressive disturbances** to account for temporal dependence in ridership while estimating the effects of protected bike lane expansion on ridership.

Project Type: Academic (Independent Study, Civil Eng. Dep. Urban Systems Engineering Lab)
Supervisor: Prof. Markus Schläpfer
Date: Sept. – Dec. 2024

Source: Transportation Alternatives, NYC Protected Bike Lane Tracker.

A Time Series Study of Bike Infrastructure Impact on System-level Citi Bike Usage in New York City

Amy Xiaoqian Chen
Fall 2024 Independent Research
Supervised by Dr.Markus Schlaepfer, Urban Systems Engineering Lab

December 31, 2024

Abstract

This study analyzes borough-level statistics on urban metrics such as daily bike trips and bike lane lengths across New York City boroughs, spanning from 2021 to September 2024. By examining travel patterns and seasonal effects on cycling, the study highlights the critical role of bike infrastructure expansion in influencing system-wide Citi Bike usage. Our findings align closely with those of Xu and Chow's 2019 study, while revealing that bike lane length has a greater impact on ridership compared to the number of active stations. Statistical analysis shows that each additional mile of bike lanes increases daily ridership by approximately 115.60 trips, underscoring the value of continued investment in sustainable urban transportation. This work emphasizes the importance of leveraging dynamic data models for informed policy design and long-term mobility planning.

1 Introduction



Figure 1: Bike Lane Expansion in NYC from 2022 to 2024 [1]

the five years from 2022 to 2026[1]. From the Transportation Alternatives data tracker, the miles of bike lane expansion is still behind the city's set target[1]. The spatial allocation of the new bike lane projects are shown in Figure 1[1].

Therefore, this study focuses on analyzing recent patterns of cycling at the borough level from 2021 to September 2024. The primary goal is to understand the latent demand for bicycles at the city level, the change in behavior and the associated environmental impacts induced by the expansion of the bicycle infrastructure in New York City. The methodology of this study is based on the article by Xu and Chow in 2019 in the International Journal of Sustainable Transportation[3]. From a city government's perspective, understanding the magnitude of causality between bike infrastructure and bikesharing ridership at the urban scale is helpful for future investment decisions. A time series model is created relating Citi Bike ridership and bike infrastructure improvement over time, to help predict cycling demand and ridership growth over the next few years using the most recent data.

2 Data Description

Citi Bike has made ridership data available to the public. On the Citi Bike website, I downloaded datasets for every month from Jan. 2021 to Sept. 2024, each file includes: ride id, rideable type, started at, ended at, start station name, start station id, end station name, end station id, start lat, start lng, end lat, end lng, and member casual. Under riderable type, there is electric bike and classic bike; under member casual, there is member and casual. I processed and aggregated the data by 1) defining borough boundaries by latitude and longitude, 2) sort and aggregate individual rides into boroughs, 3) counting and averaging total rides in each borough by week, defining the first week of 2021 as week 1. There are 196 weeks in my data. The average daily trips per week in New York City is 22292, and the average daily trips per week in Manhattan is 74722; more data are included in Table 3 in the Methodology section. The unit of week is chosen to "allow for reaction time to new bike infrastructure"[3].

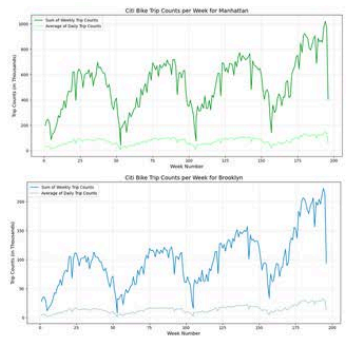


Figure 2: Citi Bike Weekly Ridership in Manhattan and Brooklyn

Figure 2 shows Citi Bike weekly ridership in Manhattan Citi Bike weekly ridership in Brooklyn, from week 1 (Jan. 2021) to week 196 (Sept. 2024). There is a clear annual pattern with nonactive bike ridership in the winter, and active bike ridership in the summer. However, it is worth noting that in 2024, both ridership in Manhattan and Brooklyn demonstrate a spike compared to previous years. Figure 3 shows the composition of total Citi Bike trip counts from 2021 to 2024. There is a clear increase of total trip counts, and a growing percentage of ridership in Brooklyn.

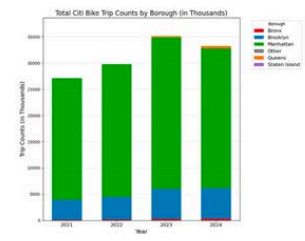


Figure 3: Total Citi Bike Trip Counts by Borough

Figure 4 compares weekday daily trip counts and weekend daily trip counts. The purple lines are generally higher than the yellow ones, indicating that the average daily ridership on weekdays is higher than on weekends.

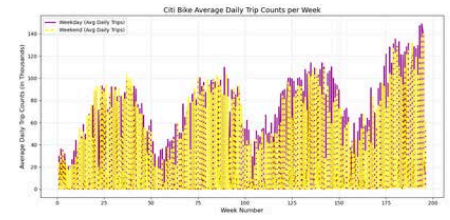


Figure 4: Citi Bike Average Daily Trip Counts per Week

Weather data are used as independent variables which intuitively effect one's decision choosing between different transit modes. Weather data is retrieved from National Climatic Data Center's station databases, specifically the New York Central Park Obs Belvedere Tower (USW0094728) weather station, from 2021 to 2024. Similar to Xu and Chow's study, the weather attributes considered are precipitation (in), snow depth (in), mean temperature (F) and mean wind speed (mph), to avoid issues of multicollinearity in the regression model.

Other independent variables included are bike lane length and the number of active stations. Using data sheets from Transportation Alternatives which summarize New York Department of Transportation (DOT) official data, I was able to plot the bike lane length (mile) change per month from 2022 to 2024 September. The starting bike lane length in 2022 is considered to be 1350 miles, calculated from data on Mayor's Office of Climate and Environmental Justice website[2]. I could not obtain bike lane length data from before 2022, nor miles per borough. However, Figure 6 shows the completed miles of bike lane in each borough from 2022 to 2024, which can be used for further study.

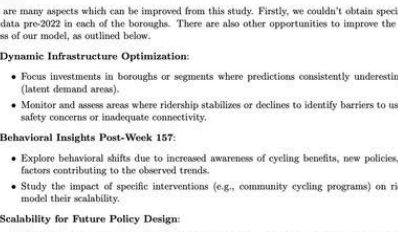


Figure 5: Total Bike Lane Length (Mile) change in New York City from 2022 to Sept. 2024

Table 1: Bike Lane Miles Completed per Borough between 2022 and 2024

Miles by Borough	2022	2023	2024	Total	% of Total
Bronx	4.2	15.07	5.48	24.75	32.4%
Brooklyn	6.07	4.64	1.19	11.90	15.6%
Manhattan	4.88	7.35	4.25	16.48	21.5%
Queens	4.70	10.62	6.85	22.17	29.0%
Staten Island	0.00	1.20	0.00	1.20	1.6%
Total	19.85	38.88	17.77	76.50	**100.0%**

In summary, the data used in this study was sourced from Citi Bike website, NYC DOT and Transportation Alternatives, Mayor's Office of Climate and Environmental Justice, and National Climate Data Center. Aggregated metrics include:

- Average daily trip counts per week (min, max, mean, std).
- Independent weather factors: precipitation, snow depth, temperature, and wind speed (min, max, mean, std).
- Built environmental factors: bike lane length, number of active stations (min, max, mean, std).

3 Methodology

Time series modeling analyzes data points recorded at regular intervals to identify trends, patterns, and temporal dependencies. Unlike standard regression, time series models account for relationships between past and present values, making them essential for dynamic systems like urban mobility. A critical prerequisite is *stationarity*, where the mean, variance, and autocorrelation remain constant over time. Non-stationary variables, such as bike lane length or active stations, require detrending to isolate short-term fluctuations. By employing stationarity tests and time series models, this study quantifies how factors like weather and infrastructure impact bike ridership, providing insights into the elasticities of ridership with respect to urban planning decisions. We first assess the stationarity of the data using the Augmented Dickey-Fuller (ADF) test, which determines whether a given time series is stationary—that is, whether its statistical properties such as mean, variance, and autocorrelation remain constant over time. Stationarity is critical for time-series regression analysis, as non-stationary data can lead to spurious results and misleading conclusions. Variables found to be non-stationary are detrended to ensure their suitability for regression analysis. The null hypothesis (H_0) for the ADF test is that the series has a unit root, meaning it is non-stationary.

Table 2: MacKinnon Approximate p-values for Variables

Variable	MacKinnon Approximate p-value
avg.daily.trips.mean	0.0098
avg.precipitation	0.0000
avg.snow	0.0001
avg.temperature	0.0000
avg.wind.speed	0.0006
FullLength	0.9608
ActiveStations	0.8451

Table 2 presents the MacKinnon Approximate p-values for each variable, indicating whether the time series is stationary. Variables with p-values below the conventional significance threshold (e.g., 0.05) are considered stationary. For example, average daily trip counts, precipitation, snow depth, temperature, and wind speed are all stationary, reflecting their consistent fluctuation around a stable mean over time. In contrast, bike lane length (`FullLength`) and active stations (`ActiveStations`) exhibit p-values of 0.9608 and 0.8451, respectively, indicating non-stationarity. This result suggests that while weather variables fluctuate around short-term seasonal patterns, infrastructure variables exhibit longer-term growth trends tied to urban development.

Table 3: Summary Statistics by Year

Variable	2022 (Weeks 54-105)	2023 (Weeks 105-157)	2024 (Weeks > 157)	Overall
avg.daily.trips.mean	0.37	0.11	0.16	0.00
avg.precipitation	0.00	0.00	0.00	0.00
avg.snow	0.00	0.00	0.00	0.00
avg.temperature	0.36	0.00	0.04	0.00
avg.wind.speed	0.00	0.06	0.35	0.00
ActiveStations	0.37	0.75	0.23	Week < 112: <0.001; Week ≥ 112: 0.003
FullLength	0.98	1.00	1.00	0.96

Table 3 provides a breakdown of summary statistics for each variable across different time periods: 2022 (Weeks 54–105), 2023 (Weeks 105–157), and 2024 (Weeks>157). The inclusion of year-specific statistics highlights how bike ridership and its associated factors evolve over time. For instance, active stations show significant differences in stationarity across time periods. In earlier weeks (Weeks<112), active stations demonstrate high statistical significance (p-value<0.001), while in later weeks (Weeks ≥112), the p-value increases to 0.003, indicating potential shifts in station usage patterns over time. Similarly, the summary statistics suggest that bike lane length becomes stationary only after detrending, indicating that long-term trends in bike lane infrastructure, likely driven by citywide urban planning efforts, are a dominant factor.

Table 4: Results of the Augmented Dickey-Fuller (ADF) Test on Detrended Data

Variable	Description	ADF Statistic	p-value
Bike Lane Length	Detrended bike lane length data	-4.079	0.001
Active Stations	Detrended active stations data	-5.221	7.97×10^{-6}

Table 4 reports the results of the ADF test after detrending non-stationary variables. The ADF statistics

- γ, α: Variance parameters.

Table 6 summarizes the characteristics of the variables used or will be used in the time series regression analysis.

Table 6: Characteristics of variables used in the time series regression analysis

Variable	Definition	Source	Min	Max	Mean	Std. Dev
Dependent Variables						
Average Daily Trip Counts	The average daily trips per week in New York City	Bikesharing system operators: Citi Bike website	0.14	145671.43	22291.98	34064.60
	The average daily trips per week in Manhattan		5682.86	145671.43	74722.40	29305.24
	The average daily trips per week in Brooklyn		1071.86	31843.43	14125.52	6581.18
	The average daily trips per week in Bronx		21	1757.14	649.89	436.52
	The average daily trips per week in Queens		0.43	2273.86	463.54	655.50
	The average daily trips per week in Staten Island		0.14	1.29	0.71	0.81
Independent Weather Factors						
Precipitation	The average precipitation per week (inch)	National Climatic Data Center	0	1.17	0.15	0.18
Snow Depth	The average depth of snowfall per week (inch)		0	2.49	0.04	0.22
Temperature	The average daily temperature per week (°F)		26.29	83.86	57.64	15.38
Wind Speed	The daily average wind speed per week (mph)		7.7	20	13.72	2.53
Build Environmental Factors						
Bike Lane Length	The total length of bike lanes/paths (mile) in five boroughs in NYC	NYC Department of Transportation	1350	1426	1384.58	26.55
	The additional length of bike lanes/paths (mile) in Manhattan		4.25	7.35	5.49	1.47
	The additional length of bike lanes/paths (mile) in Brooklyn		1.19	6.07	3.63	2.54
	The additional length of bike lanes/paths (mile) in Bronx		4.20	15.07	8.92	5.94
	The additional length of bike lanes/paths (mile) in Queens		4.70	10.62	7.06	2.74
	The additional length of bike lanes/paths (mile) in Staten Island		0.00	1.20	0.40	0.58
Active Stations	The average number of daily active stations per week in New York City	Bikesharing system operators: Citi Bike website	1217	2114	1714.78	272.4

4 Results and Discussion

Table 7 shows the results of the regression analysis for New York City. The coefficient for bike lane length ($\beta = 115.60$, p-value $= 0.002$) is positive and statistically significant, suggesting that each additional mile of bike lanes contributes approximately 115.60 trips on average. Conversely, precipitation has a significant negative effect ($\beta = -7456.83$, p-value $= 0.002$), indicating a substantial decrease in ridership during adverse weather conditions.

The regression diagnostics reinforce the robustness of the model. With an R^2 of 0.803 and Adjusted R^2 of 0.795, the model explains a large portion of the variation in daily trip counts. Additionally, the low p-value for the F-statistic confirms the overall significance of the model.

These findings highlight the critical role of infrastructure investments, particularly the expansion of bike lanes, in encouraging ridership.

Table 7: Results of the regression analysis

ARCH family regression – AR disturbances
Sample: 1–147; Number of observations = 147
Distribution: Gaussian

Model	New York City (Model 1)			
Variable	**Coefficient**	**Std. Error**	**t-statistic**	**p-value**
Dependent Variable				
Average Daily Trip Counts (NYC)				
Precipitation	-7456.83	2368.16	-3.23	0.002
Snow Depth	-132.58	2777.54	-0.05	0.962
Temperature	483.72	33.23	14.56	0.000
Wind Speed	74.11	195.46	0.38	0.705
Bike Lane Length	115.60	35.75	3.23	0.002
Active Stations	1.34	3.81	0.35	0.726
Intercept	-166400.00	43600.00	-3.81	0.000
Model Diagnostics:				
R^2	0.803			
Adjusted R^2	0.795			
F-statistic	95.11			
Prob(F-statistic)	6.79e-47			
AIC	2875.0			
BIC	2896.0			
Log-Likelihood	-1430.3			

Notes: Independent variables include weather conditions, bike lane length, and active stations. All variables are weekly averages. Model diagnostics include R^2, Adjusted R^2, F-statistic, and other metrics for model evaluation.

The results presented in this study align closely with the findings of Xu and Chow's 2019 paper. Table 8 summarizes the key similarities and differences between the two studies. However, for our study, Bike Lane Length plays a more significant role in promoting ridership than the number of active stations in the system.

Table 8: Comparison of Key Findings with Xu and Chow's 2019 Results

Variable	Our Study (01/2022- 09/2024)	Xu and Chow (07/2013-05/2016)
Precipitation	Negative and significant ($\beta = -7456.83$)	Negative and significant
Temperature	Positive and significant ($\beta = 483.72$)	Positive and significant
Wind Speed	Negative but not significant ($p = 0.705$)	Negative and significant
Bike Lane Length	Positive and significant ($\beta = 115.60$)	Positive and significant ($\beta = 102.37$); in Manhattan ($\beta = 285.09$) which we didn't get to do
Active Stations	Positive and not significant	Positive and significant ($\beta = 42.87$)

5 Conclusion

Using week 1-157 as the training sample (2022 to 2023), we were able to evaluate the robustness of our model in predicting actual ridership. The predictions starting from **week 157** (out-of-sample prediction highlight the model's ability to generalize trends in bike ridership. From week 157 onwards, there is a noticeable alignment between the observed data and predictions. However, it underestimates the ridership, with the green line of our model prediction lying under the observed trend. This indicates the influence of exogenous factors like attitude change or policy shifts that positively promote cycling.

Key Insights:

- **Sustained Growth**: The observed ridership growth aligns with infrastructure expansions in prior weeks, showcasing the long-term benefits of bike lanes.
- **Predictive Robustness**: The model's predictions from week 160 closely follow observed data, reflecting its reliability in capturing key trends, though minor variations point to the need for incorporating finer external variables.

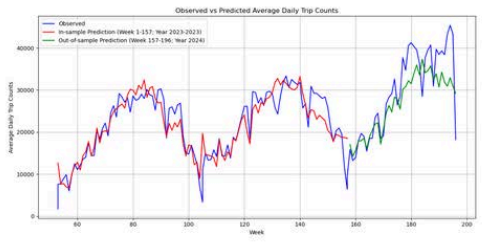


Figure 8: Observed vs. Predicted Average Daily Trip Counts (Training Sample: 2022-2023; Observed: 2023-2024 Sept.

Evaluation

There are many aspects which can be improved from this study. Firstly, we couldn't obtain specific bike lane miles data pre-2022 in each of the boroughs. There are also other opportunities to improve the comprehensiveness of our model, as outlined below.

1. **Dynamic Infrastructure Optimization**:
   - Focus investments in boroughs or segments where predictions consistently underestimate growth (latent demand areas).
   - Monitor and assess areas where ridership stabilizes or declines to identify barriers to usage, such as safety concerns or inadequate connectivity.
2. **Behavioral Insights Post-Week 157**:
   - Explore behavioral shifts due to increased awareness of cycling benefits, new policies, or cultural factors contributing to the observed trends.
   - Study the impact of specific interventions (e.g., community cycling programs) on ridership and model their scalability.
3. **Scalability for Future Policy Design**:
   - Use this predictive framework to test "what-if" scenarios, such as introducing additional miles of bike lanes or seasonal programs, and evaluate potential ridership impacts beyond week 180.
   - Monitoring the Elasticity of Ridership: As infrastructure grows, the marginal returns (e.g., additional trips per mile of bike lane) may decrease, requiring policymakers to focus on complementary factors like safety or convenience.

By refining the model and leveraging its predictive insights, urban planners can further enhance mobility strategies, ensuring equitable and efficient development of biking infrastructure across all neighborhoods. The observed trends post-week 157 reaffirm the positive correlation between bike lane expansions and ridership growth. Our statistical analysis shows that each additional mile of bike lanes contributes approximately 115.60 trips on average. This supports continued investments in sustainable urban transport systems, integrating dynamic data models for policy formulation and long-term mobility planning.

Data Sources

- Citibike Trip Data: `https://www.citibikenyc.com/system-data`
- NYC Weather Observations: `https://www.ncdc.noaa.gov/`

References

[1] Transportation Alternatives. Bike lanes project. `https://projects.transalt.org/bikelanes`, 2024. Accessed: December 30, 2024.

[2] NYC Mayor's Office of Sustainability. Bikes and pedestrians. `https://climate.cityofnewyork.us/subtopics/bikes-and-pedestrians/`, 2024. Accessed: 30 December 2024.

[3] S. J. Xu and J. Y. J. Chow. A longitudinal study of bike infrastructure impact on bikesharing system performance in new york city. *International Journal of Sustainable Transportation*, 14(11):886–902, 2019.

PART 2: URBAN REGENERATION, MASTERPLANNING & MULTIMODAL INTEGRATION

My path into planning began in architecture, learning to appreciate the design of a single building and the immediate texture of a street corner. Over time, I found myself looking beyond the orchestration of private spaces and the individual block, fascinated by how macro-level transit investments, congestion pricing, housing, and upzoning policies gradually reshape how people move, where people live and work, and the characteristics of neighborhoods.

I am drawn to **Transit-Oriented Development (TOD) that operates as a systemic catalyst to expand human opportunity**. Experiencing Hangzhou's rapid rail expansion showed me how a vast transit network can open up geographic access for students and newcomers, naturally creating new urban hubs and planned tech and innovation precincts that attract international talent. The TOD-oriented upzoning policies in Auckland raised new questions about neighborhood preservation, infrastructure demand, and housing preferences. The Interborough Express presents new opportunities for value-capture and equitable transit connectivity between Queens and Brooklyn.

Models are only as useful as their connection to the real world, and a planner must first be willing to stand on the ground and observe. Whether walking the pedestrian corridors of Barcelona's superblocks or riding transit to study urban renewal precincts in Sydney, my focus remains on understanding everyday routines, subtle friction points, and the actual needs of residents.

While the previous section focused on mining raw, unstructured data sources for unique urban insights, this section pivots to **network simulation, policy analysis and spatial interventions**—using data-driven pipelines to study, optimize, and design infrastructure systems on the ground. I aim to treat mathematical optimizations as a supportive framework to community engagement and field insights.

2.1

REIMAGINE MANHATTAN: A MATHEMATICAL MODELING APPROACH FOR SUPERBLOCK DEVELOPMENT

Project Type: Academic; Collaborative
Program: Columbia Summer Undergraduate Research Experiences in Mathematical Modeling
Supervisor: Prof. George Dragomir, Columbia Math Dep
Date: June – Aug. 2024; Ongoing
Presentations:

- Columbia Fall Research Symposium, 2024
- Jane Street Math Day, 2024

Inspired by the success of open streets programs and the anticipated implementation of congestion pricing in New York City, this project explores **superblocks as an urban strategy for traffic calming, safety, public space and small business revitalization** in Manhattan. Vision Zero cannot be achieved without taking sidewalk and roadway design seriously, especially at large intersections that pose persistent risks for cyclists and pedestrians.

Using a graph-theoretic model that integrates street network structure with Point-of-Interest data, the project evaluates superblock suitability across Manhattan and simulates a **seeded growth strategy for pilot implementation**. The methodology offers a data-driven pathway for superblock development.

Role: Led project definition and methodology design; conducted suitability score modeling and coding, produced analytical visualizations, carried out independent fieldwork, designing ArcGIS Storymap, and continuing the research write-up and storymap design process.

Research Abstract

Reimagining Manhattan: A Mathematical Modeling Approach to Superblock Development
CSUREMM 2024 Abstract

Shiva Yeshlur, Amy Xiaoqian Chen, Cionnie Pineda

July 2024

Abstract

Introducing superblocks as an urban design and traffic planning strategy to Barcelona's grid structure has radically transformed the city's neighborhood fabric. Leveraging the creation of new pedestrian and cyclist-oriented inner-block zones has alleviated air and noise pollution, congestion, and a host of other environmental concerns. Current research often falls into one of two categories: urban planning-focused qualitative research on superblocks real-world implementation, or graph optimization and network theory-based quantitative models for superblock partitioning.

The former utilizes urban planning expertise and provides far more insight into region-specific governance and regulation of superblocks but is usually far less helpful to the study of superblock development in other urban regions. The latter, while desirable for simulating the impact of superblock transformations on traffic networks, largely ignores inherent community structures that make certain algorithmic choices for superblock transformation unfeasible.

Our novel approach relies on a graph-theoretic model that incorporates Point-of-Interest (POI) data into a multi-level, hierarchical partitioning strategy using Leiden Community Detection. We transform the Manhattan grid into a weighted graph representation with nodes as intersections and edges as road segments between intersections, utilizing POI scores from feature-count data from Open Street Maps (OSM) together with edge lengths to create a geospatially- and POI-aware clustering model. We then simulate a potential growth strategy of superblocks across Manhattan by taking advantage of the hierarchical partitioning strategy to generate a geographically rich set of initial superblocks, dubbed our "pilot program."

1

Points of Interest Categories

Food, Shops, & Entertainment		Arts, Culture, & Tourism		Education, Work, & Care		Space Openness, Comfort & Flexibility	
Building	retail	Building	cathedral	Building	apartments col 3	Building	pavilion
Building	cinema	Building	terrace	Building	school	Amenity	bicycle_parking
amenity	restaurant	Building	public	Building	residential	amenity	bench
amenity	café	Building	church	Building	dormitory	amenity	fountain
amenity	bar	Building	synagogue	Building	commercial	highway	cycleway
amenity	pub	Building	museum	Building	kindergarten	highway	footway
amenity	fast_food	Building	civic	Building	university	highway	bus_stop
amenity	cinema	Building	hotel	Building	office	landuse	grass
amenity	nighclub			Building	college	landuse	recreation_ground
amenity	food_court	Building	theatre	amenity	social_facility	landuse	greenfield
leisure	dance	Building	bridge	amenity	pharmacy	landuse	flowerbed
leisure	ice_rink	amenity	place_of_worship	amenity	community_centre	leisure	park
leisure	social_club	amenity	events_venue	amenity	university	leisure	playground
leisure	bowling_alley	amenity	theatre	amenity	social_centre	leisure	picnic_table
		amenity	music_venue	amenity	library	leisure	garden
		amenity	arts_centre	landuse	residential	leisure	dog_park
		historic	memorial	leisure	fitness-centre	leisure	outdoor_seating
		historic	building	leisure	schoolyard	natural	tree
		historic	district			natural	water
		historic	monument			natural	tree_row
		historic	heritage			railway	station
		tourism	attraction			tourism	picnic_site
		tourism	hostel				
		tourism	gallery				
		tourism	hotel				
		tourism	artwork				
		tourism	museum				
		tourism	viewpoint				

SUPERBLOCKS FOR URBAN COMMUNITIES

SUPERBLOCKS FOR URBAN COMMUNITIES

Reimagining Manhattan: A Mathematical Approach to Superblock Development for Enhanced Travel Safety, Leisure, and Community Living

Amy Xiaoqian Chen, Shiva Yeshlur, Cionnie Pineda | Columbia Math Department
July 23, 2024

Reimagining Manhattan: A Mathematical Modeling Approach to Superblock Development

Amy Xiaoqian Chen, Shiva Yeshlur, Cionnie Pineda
Columbia Summer Undergraduate Experiences in Mathematical Modeling (Columbia University)
Mentored by Professor George Dragomir and Vihan Pandey

ArcGIS StoryMap
for full pilot program information and growth strategy animation

Contact: xc2671@columbia.edu

PART 2: URBAN REGENERATION, MASTERPLANNING & MULTIMODAL INTEGRATION 2.1

01 Introduction

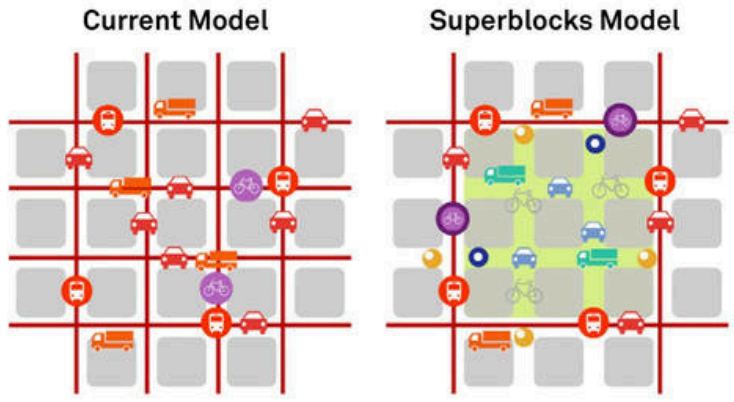


Figure 1 : Superblock model diagram (Eggiman 2022) [1]

Objective and Aim:

- The project aims to develop a proposal for integrating superblocks into Manhattan's urban street layout by:
  1) Developing a model for Superblock Suitability Score;
  2) Demonstrating a growth strategy to pilot and connect superblocks for a full urban layout transformation.

How can we design a superblock program for Manhattan that prioritizes communities and enhances urban life?

Superblock and POI Definition

- A city block, residential block, or urban block, refers to the smallest collection of buildings enclosed by streets.
- Superblocks involve grouping several city blocks, restricting or pacifying through traffic on interior streets, and transforming these spaces into bike lanes, green corridors, and public spaces.
- Points of Interest (POI): locations of objects in urban areas, such as cafés, galleries, and parks, which influence superblock suitability.
- POI Categories: Catering, Shops & Entertainment; Arts, Culture & Tourism; Education, Work & Care; Space Openness & Comfort; Penalty

02 Data Collection and Processing

- Turning Manhattan into a graph of nodes, representing street intersections, and edges, representing streets, with OSMnx [2] .
- Using data from Open Street Maps (OSM), **87 POI features** are counted in **400m radii** around each of the **4594 nodes** in Manhattan (400m= approx. a five-min walking time).
  - Nodes without features in the fixed radii are indicated with "0" value counts in all features, encouraging immediacy of POI feature distributions.
- Features falling under the same POI category were first min-max normalized, added, and normalized again to reflect the category score.

03 Methodology

POI Node-to-Edge Weighting

- Normalized category scores are utilized to compute final Superblock Suitability Score, with given alpha and beta parameters, and only considering penalties if they are above the mean penalty score:

$$s(n_i) = \alpha||w_i||_\infty + \beta_{i_1} w_{i_1} + \beta_{i_2} w_{i_2} + \beta_{i_3} w_{i_3} + \beta_{i_4} w_{i_4} - \beta_{5_i} p_i$$
$$p_i \neq 0 \iff p_i > \bar{p}$$

- Graph edges use node scores and lengths, passed through a sigmoid-like function, for a geospatially-aware weighting function:

$$f(e_{ij}) = \frac{1}{1 + e^{-l + \frac{s(n_i)+s(n_j)}{2}}}$$

Multilevel Leiden Clustering

- The Leiden Community Detection Algorithm [3] is used on two levels:
  - Level 1: Processes geographic regions of Manhattan, to incentivize seeded growth strategy of superblock development.
  - Level 2: Creates superblocks based on densely connected sub-communities, utilizing POI information.
- Algorithm optimizes Graph modularity, a measure of edge density and edge distributions across each community cluster:

$$Q = \frac{1}{2m} \sum_{i,j} \left[f(e_{ij}) - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

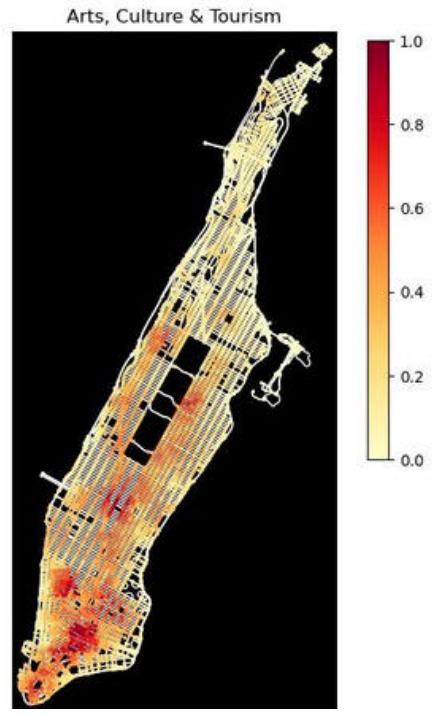
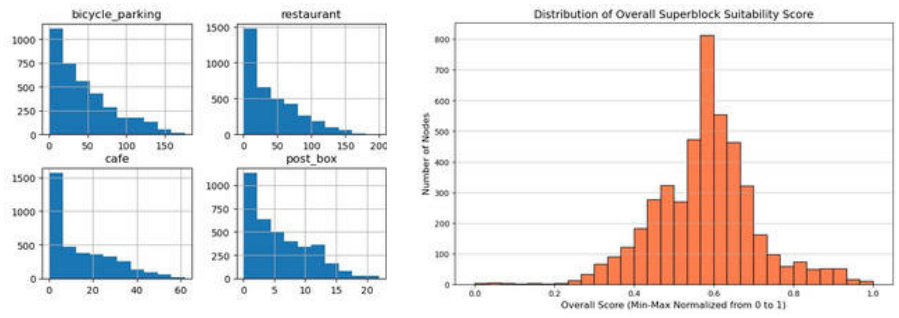


Fig 4.2 - normalized POI scores for the "Arts, Culture, Tourism" category. For non-historic POI, Lower Manhattan receives significantly higher scores. Higher score= higher interest

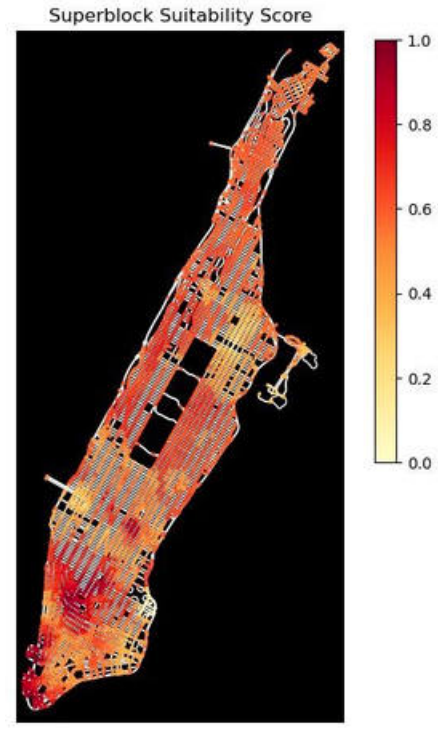
04 Results and Analysis

Superblock Suitability Score Normalization

- After processing each of our 4 POI categories and additional penalty category (i.e. parking lots) and normalizing our scores, a normal distribution of final suitability scores is recovered:



Figures 4-1 : (Left) Value-count distribution of example features in POI categories, and (Right) Distribution of Final Superblock Suitability Scores. Despite count-value data having long tailed-distributions, our final weighted score recovers a normal distribution of Suitability across geographic locations in Manhattan.



Figures 4-3 : Overall Weighted Superblock Suitability Score. Upon score weighting, Suitability Scores are more evenly distributed across Manhattan.

05 Implementation

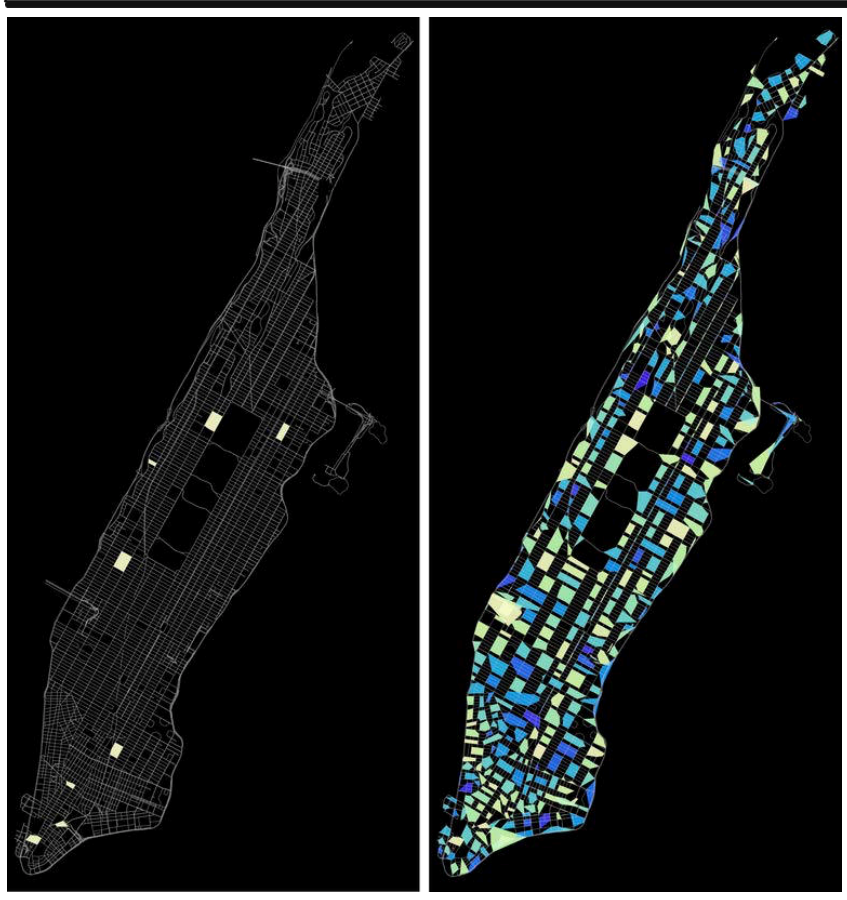
Seeded Growth Strategy and Pilot Programs

- To demonstrate potential urban planning development of superblocks over time, a seeded growth strategy was developed to encourage geographically- and characteristically-diverse developments across Manhattan.
  - Utilizes first level of Multilevel strategy to generate geographic regions, while the second layer encodes high-density subregions in the regional context.
- Pilot superblocks (first round of development) of particular interest to urban planners to research, design, implement, and assess initial transformations of Manhattan city blocks.

06 Next Steps

Optimizing for Global Suitability

- Choices for suitability score calculation parameters and edge weighting function are largely heuristic;
  - more experimentation is needed to fully understand how edge weight transformations modify the final selection of pilot-program superblocks.



Next Steps (continues)

- Leiden Algorithm approach also retains many individual parameters for community detection;
  - Need to develop a way to integrate optimization choices with built-in Leiden model parameters.

Assessing Pilot Program and Growth Strategies

- How do certain pilot program choices motivate future superblock development dynamically?
- How do different growth strategies improve/exacerbate air pollution, congestion, and larger urban sustainability problems?

Figure 5-1: (Left) Potential Pilot Program Development and (Right) Fully-Realized Development based on a Seeded Growth Strategy.

References: [1] S. Eggimann. "The potential of implementing superblocks for multifunctional street use in cities." In: Nat Sustain (2022). DOI: https://doi.org/10.1038/s41893-022-00855-2. [2] G. Boeing. "Modeling and Analyzing Urban Networks and Amenities with OSMnx. Working Paper". In: (2024). URL: https://geoffboeing.com/publications/osmnx-paper/. [3] V. A. Traag, L. Waltman, and N. J. van Eck. "From Louvain to Leiden: Guaranteeing well-connected communities." In: Scientific Reports, 9(1), 5233. (2019). DOI: https://doi.org/10.1038/s41598-019-41695-z.

DEPARTMENT OF MATHEMATICS
COLUMBIA UNIVERSITY IN THE CITY OF NEW YORK









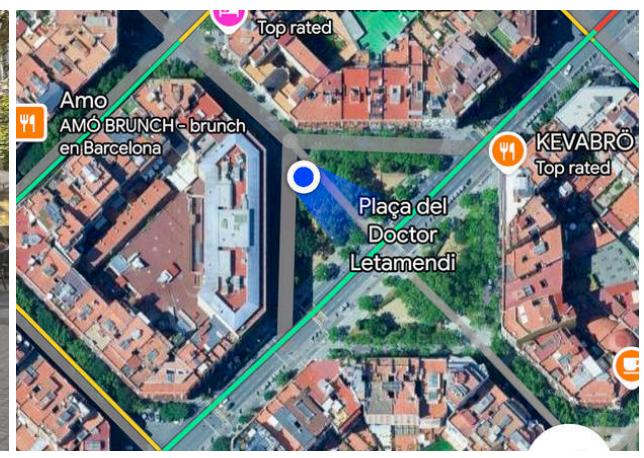
Manhattan

Conducted fieldwork at proposed pilot superblock sites and along streets that are part of the Open Streets program in New York City.

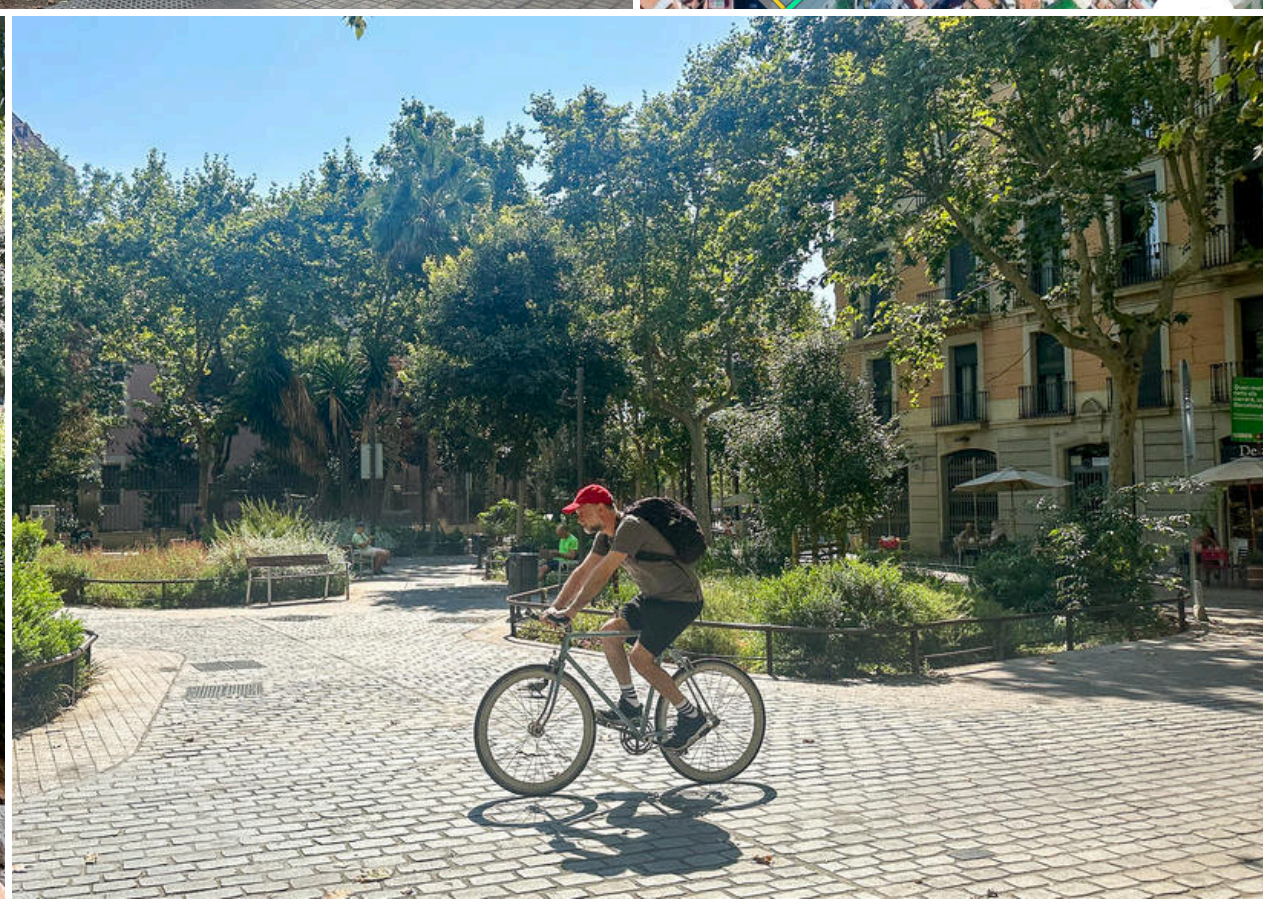
In August 2025, I conducted independent fieldwork in Barcelona, garnering feedback from the locals and observing how the implementation actually works: for example, what happens if a delivery truck or car has to pass through an interior street in a superblock neighborhood? I also documented different superblock typologies and ecological practices. This experience in Barcelona will prepare me to extend our math modeling methodology and make more grounded suggestions to the New York City planning commission/BIDs as I train to become a planner.

Barcelona









Confidential work

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Confidential work

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PART 3: CLIMATE RISK, ASSET ANALYTICS & SOCIOECONOMIC POLICY

Protecting vulnerable neighborhoods and critical municipal assets requires a careful understanding of how environmental hazards, economic realities, and long-term public policies interact on the ground.

This section bridges **applied microeconomics methods with coastal resilience planning** to study how physical, financial, and social structures perform under stress. By analyzing the intersection of 3D coastal flood statistics and real estate market valuations in San Francisco, these case studies evaluate the real-world boundaries and patterns of climate vulnerability. This framework is deeply informed by continuous empirical research into micro-mobility behavior in New York City, using granular Citi Bike trip records as a natural experiment to test how everyday commuting patterns adapt during extreme climate shocks like the 2023 wildfire haze event.

Extending this systems-level focus to regional resource networks, **my long-term goal is to evaluate the environmental, computational, and socio-political parameters of clean energy systems**. I am particularly interested in how we can use complexity science to model the invisible, high-stakes relationship between regional energy grids and the built environment, simulating how smart cities can balance electric transit networks with the intensive power and land demands of emerging data center infrastructure. I hope to contribute to building sustainable and equitable visions of the future.

3.1

CLIMATE RESILIENCY MODELING: MULTI-MODAL NETWORK DISRUPTION ANALYTICS UNDER EXTREME WEATHER EVENTS

Project Type: Academic; Collaborative
Course: Sustainable Urban Systems Engineering
Supervisor: Prof. Markus Schläpfer
Date: Feb. – May 2025

During the June 2023 wildfire haze event, news coverage suggested that despite public health advisories, many residents and essential workers continued to travel. This project **treats the event as a natural experiment** and tests those claims against observed mobility data, examining whether and how urban travel patterns actually changed across modes and neighborhoods.

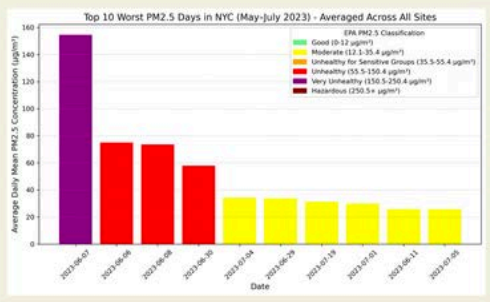
The study relies on bikeshare, subway, bus, and taxi data; air quality and weather data; and neighborhood socioeconomic indicators. **Mobility data were aggregated to the neighborhood level** and aligned with daily AQI and $PM_{2.5}$ measures to identify haze periods. The analysis combines multi-year temporal comparisons with regression modeling to isolate the effects of hazardous air quality while controlling for seasonality and weather.

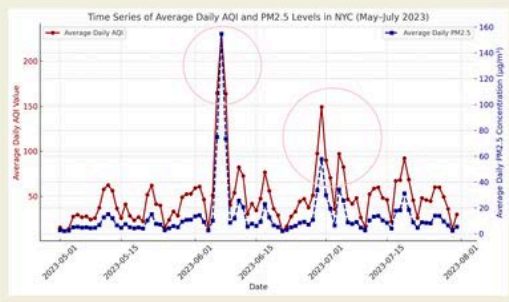
For Citi Bike, **origin–destination matrices were constructed and visualized as both absolute and percentage changes in daily ridership**, with animated OD flows used to illustrate how spatial mobility patterns shifted during haze days.

Role: Led project concept and methodology; implemented the Citi Bike analysis in Python, including large-scale data processing, statistical modeling, and origin–destination visualizations.

Data

Air Quality Ratings (AQI)



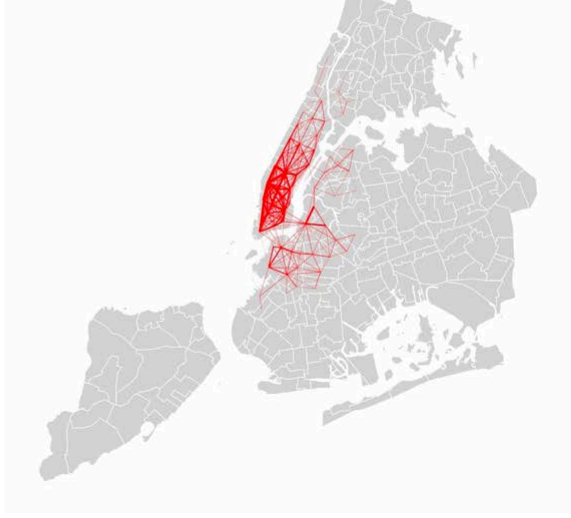


- The Air Quality Index (AQI) is a standardized scale used to communicate how polluted the air currently is or how polluted it is forecast to become.
- AQI ≤100 is generally considered satisfactory; >100 becomes unhealthy, first for sensitive groups, then for all.
- **AQI does not account for hazardous air pollutants, extreme heat, or other outdoor air risks.**

Preliminary Data Analysis | CitiBike

Dataset #1: Citibike Data

- Data: **CitiBike Data in June and July for 2022, 2023, and 2024**; spatially joined with **Neighborhood Tabulation Area** boundaries.
- Using the starting and ending longitudes and latitudes for every trip, we aggregated the daily ridership going in and out of each neighborhood, every day from June 1st to July 31st.
- **The animation on the right shows the change in daily Origin-Destination flows, with the AQI value as a reference to when the wildfire haze event happened.**



Preliminary Data Analysis | CitiBike

June 7th

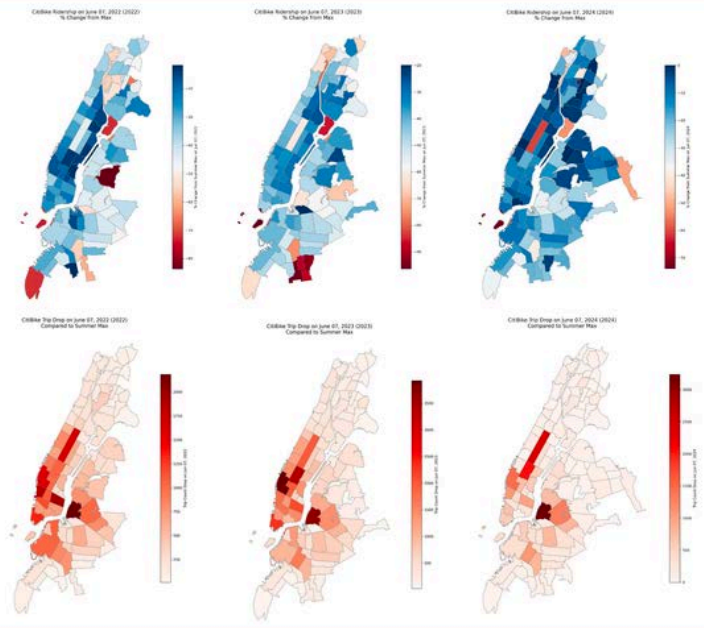
Percentage change from maximum daily ridership within each neighborhood (2022, 2023, 2024)

Darker red —> higher percentage change (more sensitivity to haze event)

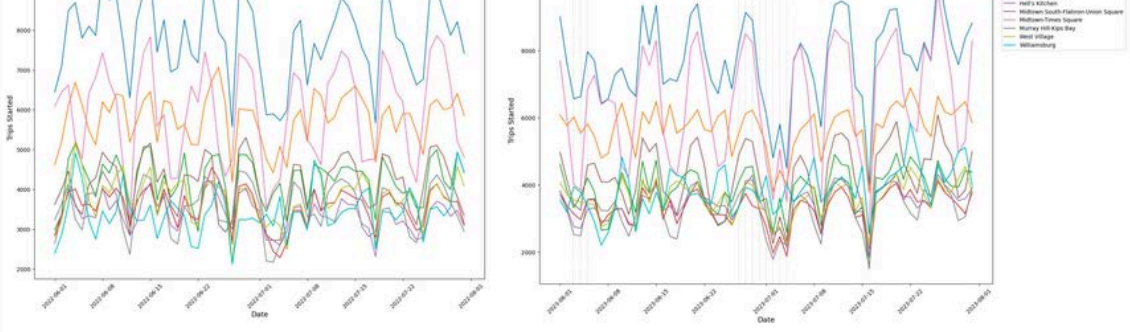
Darker blue —> lower percentage change

Drop in trip count from maximum daily ridership within each neighborhood (2022, 2023, 2024)

Darker red —> higher drop in trip counts (more sensitivity to haze event)

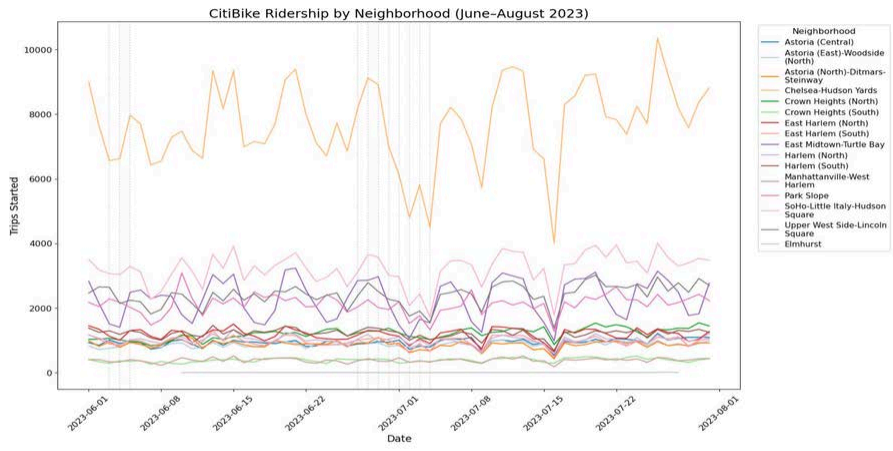


Preliminary Data Analysis | CitiBike



- Compared with 2022 and 2024 trends, the 2023 ridership from June to August 1st display a sharper decline in July, and a dip around June 2nd to June 7th, which correlates with the first haze event period.
- Even after normalizing the data by calculating the percentage drop from the highest daily ridership count for each day in each neighborhood, the line plot doesn't allow us to quantify or separate the impact of the haze event.

Preliminary Data Analysis | CitiBike



- The overall ridership trend for these selected neighborhoods is representative of the one for all New York City neighborhoods. There is a significant dip in daily ridership from late June to July 4th.
- **July 4th is a national holiday, so ridership is expected to decrease regardless of the haze event.**

Income Comparisons

Dataset #1: Citibike Data | Regression Model 3, Multiyear Panel with Weather and Year Controls

Table 3: OLS Regression Results on CitiBike Trips with Weather and Income Interaction

Model: OLS Regression with Weather Effects
Dependent Variable: Daily Trip Counts per Neighborhood

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Intercept	1506.62	317.45	4.75	0.000***
Year = 2023	152.73	84.01	1.82	0.069*
Year = 2024	379.03	49.53	7.65	0.000***
Median Household Income (centered)	0.0189	0.0005	38.88	0.000***
Precipitation (inches)	-389.95	106.02	-3.68	0.000***
Average Temperature (°F)	-3.54	4.28	-0.83	0.409
Haze Event	-209.53	88.78	-2.36	0.018**
Haze × Income (centered)	-0.0050	0.0010	-5.17	0.000***
Model Diagnostics:				
R^2	0.279			
Adjusted R^2	0.278			
F-statistic	267.5			
Log-Likelihood	-41959			
AIC	83930			
BIC	83990			

- Once again, **haze events continue to suppress ridership significantly—by about 210 trips/day.** The income × haze interaction coefficient is still negative but close to zero, rejecting the hypothesis that wealthier neighborhoods show stronger behavioral responses to air quality events.
- As expected, **precipitation strongly reduces ridership, but temperature is not significant in this model**—perhaps due to most summer days falling within a comfortable range.

3.2

STORYTELLING FOR COASTAL RESLIENCE: STATEN ISLAND NORTH SHORE AND THE MERC CORRIDOR

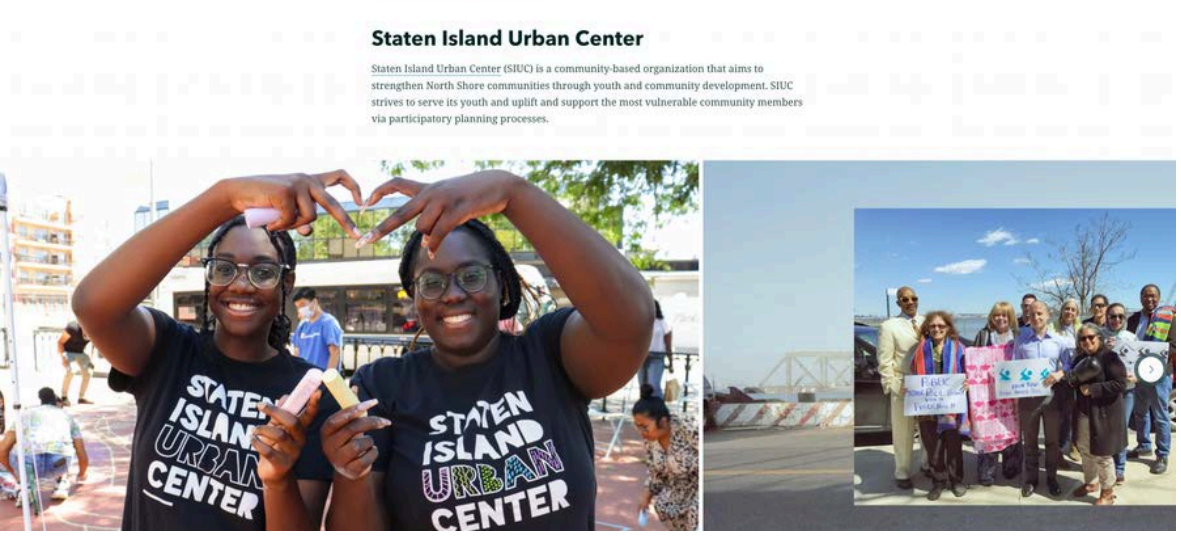
Project Type: Academic; Collaborative
Site: Staten Island North Shore, New York
Course: Building Climate Justice: Co-Creative Coastal Resilience Planning
Date: Oct. – Dec. 2023
Presentations:

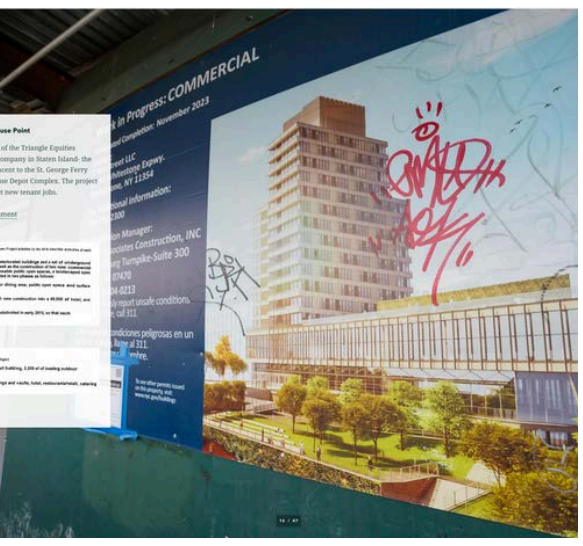
- New York Climate Justice Summit, 2024
- CUNY Columbia Environmental Justice Summit, 2024

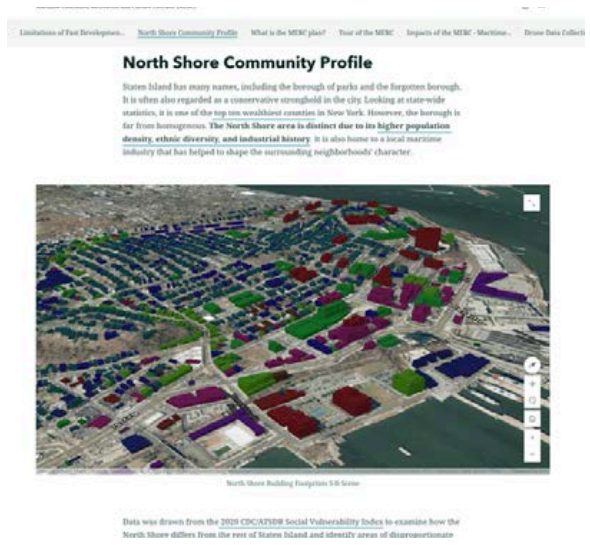
The MERC Corridor project uses both secondary data and primary data collection to reveal **environmental and social conditions on Staten Island's North Shore that are often hidden by census averages**. The analysis uses Census demographics, NYC street and building data, community air survey results, and education indicators to find patterns of environmental exposure and vulnerability.

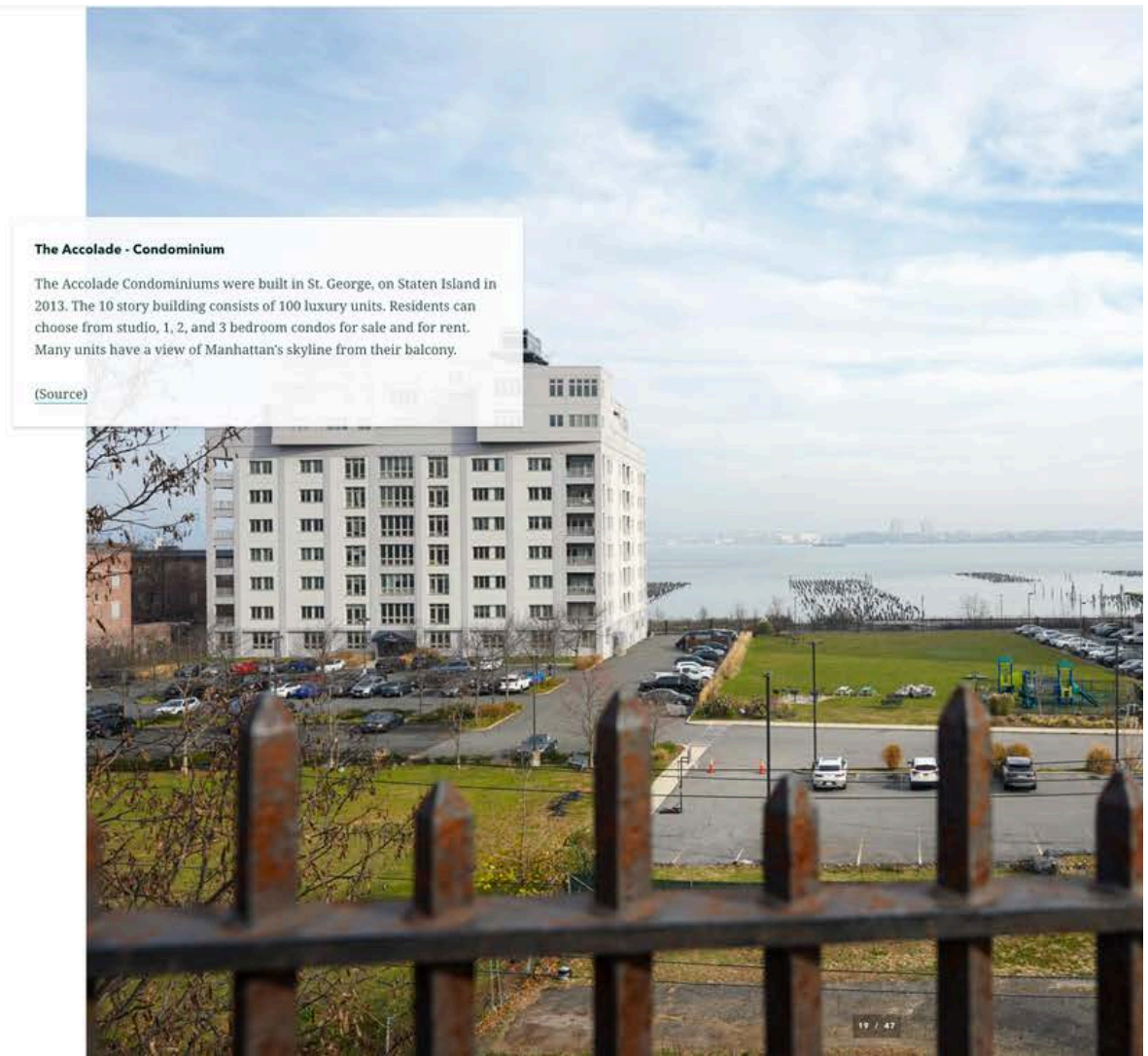
The project focuses on **co-creating knowledge with the community through citizen science tools**, such as Atmotube Pro air quality sensors and NYPD-approved drone flights. These tools produced detailed point clouds and 3D site models, helping the community build long-term advocacy capacity.

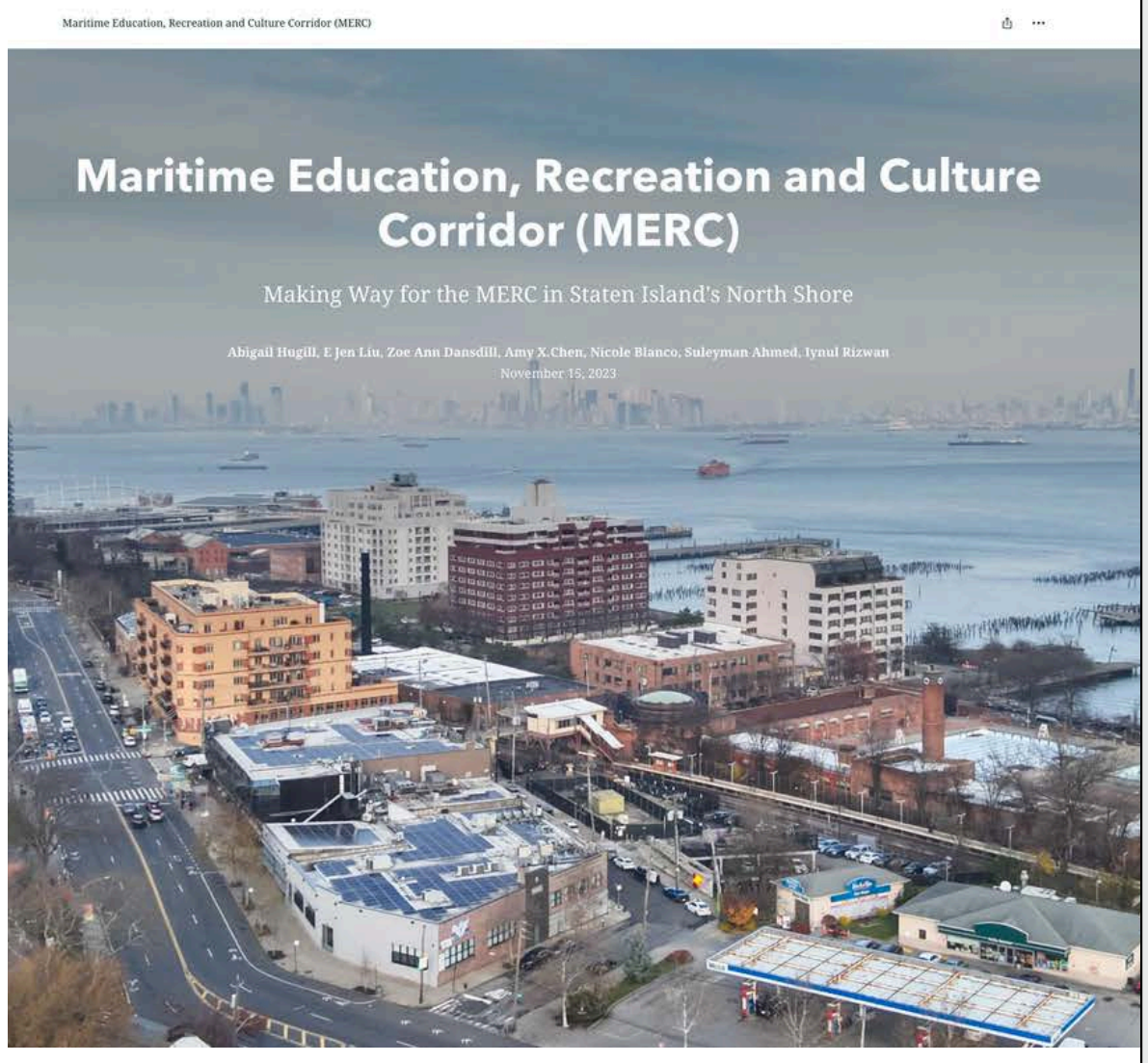
Role: We worked with the Staten Island Urban Center to define the research question through community conversations and site visits along the North Shore. I carried out fieldwork and visual documentation, helped with primary data collection, and created narrative and visual materials, including an interactive MERC corridor tour, to make technical analysis more accessible and community-driven.

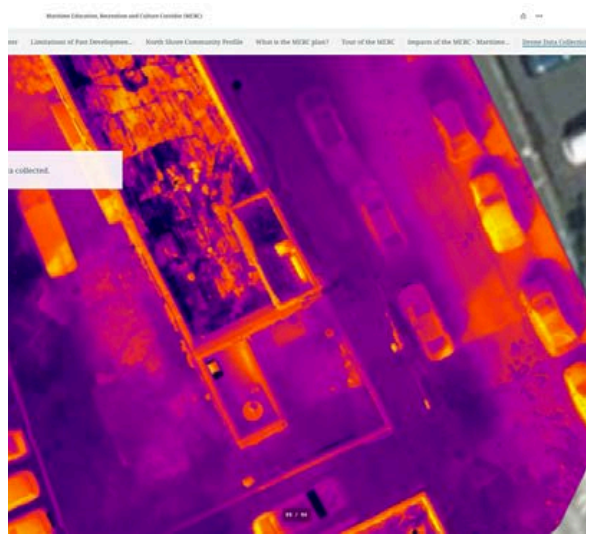


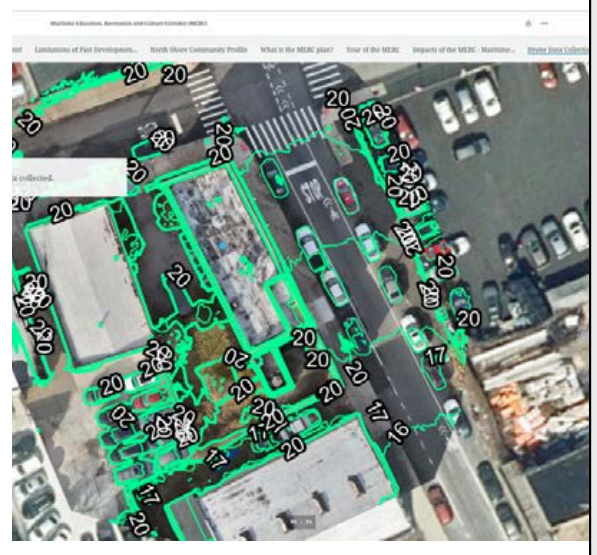


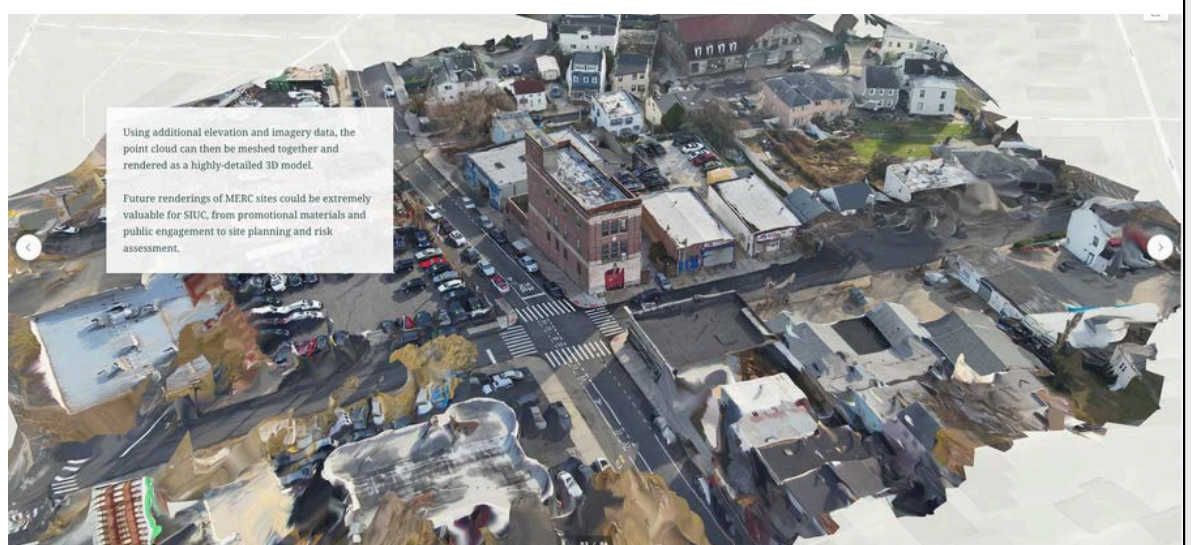












Building Climate Justice: Co-Creative Coastal Resilience Planning

Taught by Professor Paul Gallay, Kytt MacManus, and Greg Yetman; TA: Annika Nina Yu Community Partner Organizations: South Bronx Unite, Staten Island Urban Center Student Researchers: Amy Xiaoqian Chen, E Jen Liu, Zoe Ann Dansdill, Suleyman Ahmed, Iynul Rizwan, Abigail Hugill, Nicole Blanco, Ananya Pal, Alonso Valesquez, Carmelli Leal, Charitie Ropati, Daniella Donzelli Scorza, Elena Stites, Olivia Schmitt

Contact: xc2671@columbia.edu

(Mapping the Air 2023)

(Making Way for the MERC 2023)

01 Introduction

Objective and Aim:

- This course educates students and supports effective coastal resilience planning and climate justice, through social and data science learning and data acquisition and analysis, making use of emerging technologies and best practices for collaboration with environmental and climate justice practitioners. Instruction is provided in two areas:
- ○ Climate adaptation planning & climate justice, Data
- ○ science: acquisition, analysis and visualization.

How can academia and frontline communities co-produce knowledge to build climate justice in coastal resilience planning?

02 Data Collection and Processing

Co-Creation of Knowledge

- Students and instructors worked with participating community- based climate and environmental justice organizations South Bronx Unite (SBU) and Staten Island Urban Center (SIUC) to collect and analyze biological, geographic, and socio-economic data relevant to local resilience needs. The dialogue includes zoom calls, office visits, and field research.

Existing Geospatial Data

- The SBU group uses a combination of Census Tract Data, NYC Street Map Data, NYC Building Footprints, Community Air Survey Particulate Matter Data, and Truck Traffic Routes data to identify air pollution locations and calculate the aspect ratios of buildings.
- The SIUC group uses Census Tract Data, education data.

Primary Data Collection with Citizen Science Technologies

- Atmotube Pro: portable device for air quality data collection
- Drone Data Collection: With several pre-programmed missions, all drone imagery was captured over a few hours on the afternoon of Saturday, December 2nd.
  - ○ The drone captures thousands of individual data points about the distance, color, and thermal characteristics of objects within about 250 feet of the firehouse, creating a 'point cloud.' Using additional elevation and imagery data, the point cloud can then be meshed together and rendered as a highly-detailed 3D model.
  - ○ The group received approval from NYPD for the drone mission.



Figure 1 : Atmotube Pro (Mapping the Air 2023)

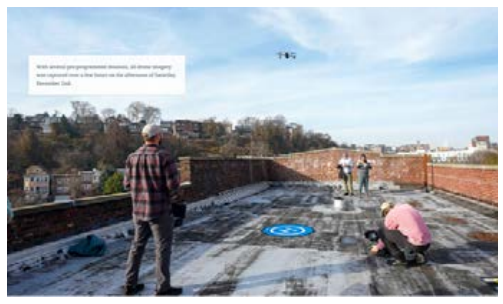


Figure 2 : Drone Data Collection in Staten Island North Shore (Making Way for the MERC 2023)

05 Reflections

True partnership and co-planning relies on trust

- Rather than an academic institution or government agency coming in with a predetermined agenda to a community-based organization, Trust is only fostered when acting with transparency, consistency, and the willingness to self-evaluate and improve practices.

Role of the academic institution in climate justice

- Academic institutions have tremendous ability and responsibility to educate students on what it means to become authors of data and change-makers. Academic institutions can engage more deeply with frontline communities through a variety of creative mediums to advance climate justice.

03 Methodology

Mapping the Air: Illustrating Air Pollution Burden and Providing Recommendations for its Measurement in the South Bronx

Profile and Route Design

- South Bronx Unite group identified locations where high traffic of people and emitting vehicles intersect. Three representative profiles – adults, children, and elders – are designed to highlight the vulnerability of different demographic groups to air pollution in the South Bronx. Air quality

data are collected from different times of day .

Street Canyons Model for Traffic-Related Pollution

- Natural ventilation is restricted in street canyons due to the flow fields not entering into the canyon entirely, trapping pollutants in the lower parts.
- Streets are split at intersections to create block by block segments. The mean height of the buildings is divided by its street width to find the segment's aspect ratio. Higher aspect ratios have a higher likelihood for poor ventilation.
  However, lower aspect ratios increase in pollutant
- accumulation as the length of the street canyon increases.

Maritime Education, Recreation and Culture Corridor (MERC): Making Way for the MERC in Staten Island's North Shore

Exploratory Regression Analyses

- Regression Analyses were ran on four areas of the Social Vulenrability Index for better understanding: Socioeconomic Status, Household Characteristics, Racial and Ethnic Minority Status, and Housing Type and Transportation

Neighborhood Profile and Aerial Imagery .

- Aerial imagery captured at 30 Hannah St offers a promising proof-of-concept for refining the MERC and promoting stronger engagement. Furthermore, all aerial imagery captured will belong to SIUC, which further grows community capacity long-term. The full interactive data sets can be found here (via Site Scan) .

Impact Modeling with Focus on Maritime Education

- The group estimates the impact that the MERC would have by looking at the number of people within walking distance of each of the proposed facilities, with a focus on maritime education.

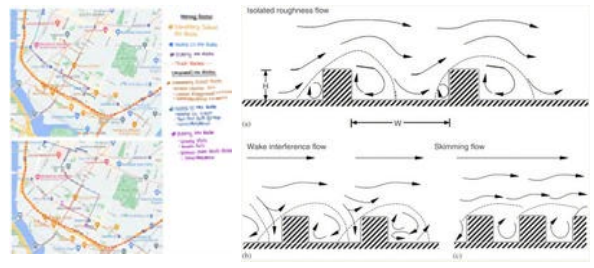
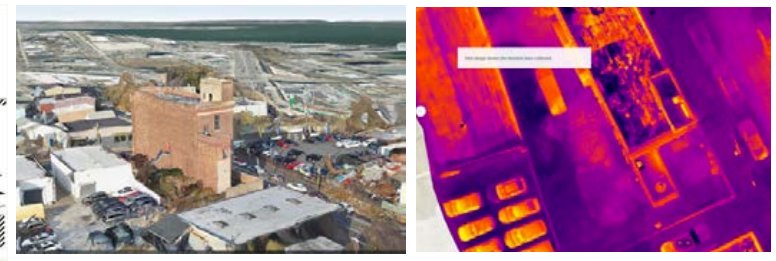


Figure 3: (left) Three Routes, and (right) Street Canyons Model



Figures 4 : (Left) 3D mesh model of waterfront, and (Right) thermal data of the waterfront collected with drone

04 Results and Analysis

Five Recommendations for Monitoring Areas

- St Ann's Avenue Between 156th and 159th St
- 162nd St Between Teller & Morris Ave Brook
- Ave Between 137th & 138th 139th St
- Between St Ann's & Cypress Ave 149th St
- Between Park & Morris Ave

Connecting Air Pollution Burden and the Need for Waterfront Access

- The transformative potential of introducing open green spaces to enhance the overall quality of life becomes glaringly apparent when considering the adverse impact on the community due to the current waterfront being monopolized by facilities like the FreshDirect warehouse, waste transfer stations, and peak power plants.

Demographics: North Shore vs. Staten Island

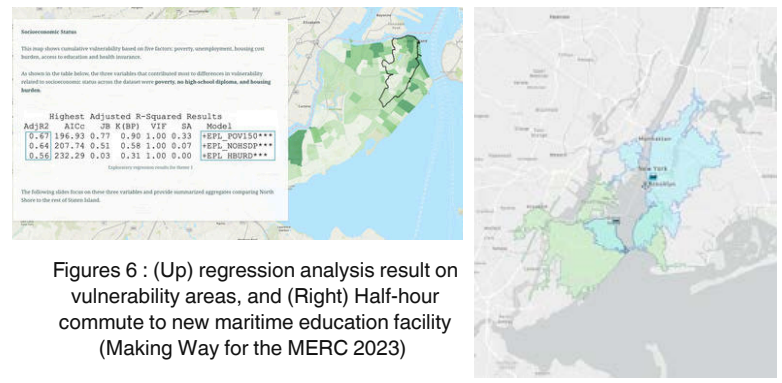
- Across multiple Social Vulnerability Index themes, there is a discernible trend of heightened vulnerability in the North Shore when compared to the Southern and Central areas of the Island. Analysis shows that the greatest disparities between North Shore and the
- rest of Staten Island are related to housing type and transportation, as well as racial/ethnic makeup.

Maritime Education

- To make maritime education accessible for the whole North Shore (and parts of Brooklyn) would be to build a maritime education complex at Fort Wadsworth, an integral part of the MERC plan. The new school would place an additional 730,636
- people within a half-hour commute to a maritime education facility (shaded in green).



Figures 5 : Bronx Aspect Ratio with QuantSensors and Recommendation Sites (Mapping the Air 2023)



Figures 6 : (Up) regression analysis result on vulnerability areas, and (Right) Half-hour commute to new maritime education facility (Making Way for the MERC 2023)

References: (Mapping the Air 2023) https://storymaps.arcgis.com/stories/7e1aec72c9ec426ca5cccd60f0bcda81; (Making Way for the MERC 2023) https://storymaps.arcgis.com/stories/4606581b7a8142d1be9e668b5838e64f

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PART 3: CLIMATE RISK, ASSET ANALYTICS & SOCIOECONOMIC POLICY 3.2

3.3

TRUCKS, FUMES, AND GROCERIES: Environmental Justice in Mott Haven–Port Morris

This project traces how truck traffic, industrial land use, and logistics infrastructure shape **environmental tradeoffs and public health outcomes in Mott Haven–Port Morris in the South Bronx**. Combining spatial analysis and historical research, I write about how highway siting, warehouses, and distribution centers concentrate diesel exhaust near homes, schools, and public housing. I created maps and collages to tell the **story of environmental sacrifice and community resilience**.

Project Type: Academic & Personal
Date: 2023; 2025
Published in Barnard Columbia Urban Review

TRUCKS, FUMES, AND GROCERIES: A TALE OF ENVIRONMENTAL JUSTICE — TRACING TRUCK EXHAUST IN MOTT HAVEN-PORT MORRIS, SOUTH BRONX, NEW YORK

BY AMY XIAOQIAN CHEN
Columbia College '26

Low-income and Black, Indigenous, and other People of Color (BIPOC) communities have historically been alienated into sacrifice zones. A sacrifice zone is a geographical area permanently altered by economic disinvestment and disproportionate environmental burden. The Mott Haven-Port Morris neighborhood in the South Bronx is a sacrifice zone. Alongside residential units and schools are three major expressways, numerous manufacturing factories, multiple waste transfer stations, and four gas-burning "Peaker" power plants. These physical infrastructures in the built environment send multiple signs of environmental racism when we trace the historical narrative behind their siting. The intersection of highway infrastructure in the neighborhood creates an epicenter for diesel truck traffic, producing constant air and noise pollution that increases asthma rates and decreases quality of life. Notably, the circulation of food driven by e-grocer FreshDirect has amplified truck exhaust in recent years. This paper intends to illustrate the lived experience of the Mott Haven-Port Morris people, using different data types to tell a real story of environmental racism and injustice right next to our Columbia campus.

Demographically, Mott Haven-Port Morris neighborhood is a 98% BIPOC community. Around the highway intersection, Census Tract 19 circling the South Bronx boundary has a 53.39% Hispanic or Latino population and 37.42% Black or African American population. Census Tract 27.01 has 69.54% Hispanic or Latino population and 31.37% Black or African American population based on the 2017 American Community Survey data (5-Year Estimates).

INDUSTRIAL SITING AND AIR POLLUTION

Now we can start unraveling the first piece of the tale: industrial siting and air pollution. The radial area highlighted in figures 2 and 3 shows the Mott Haven-Port Morris neighborhood area I will focus on. The area contains the intersection of three major traffic infrastructures: the horizontally crossing Major Deegan Expressway, the Robert F. Kennedy Bridge to Randall's Island, and the Bruckner Expressway going up north. Major pollution sites are illustrated in orange in figure 1, while major social housing and schools in the area are highlighted in cyan in figure 3, demonstrating their disturbing proximity.

The FreshDirect warehouse occupies an area of 11 football fields and has a state-of-the-art facility to generate high grocery delivery capacity and efficiency. As shown in figure 1, the warehouse sits adjacent to the highway intersection and other diesel truck-intensive businesses, such as the FoodFest Depot and the FedEx Ship Center, rendering the



Figure 1: Annotated map of a part of the Mott Haven-Port Morris neighborhood. Features three major traffic infrastructure: the horizontally crossing Major Deegan Expressway, the Robert F. Kennedy Bridge to Randall's Island, and the Bruckner Expressway going up north. Major polluting sites including the FreshDirect warehouse, FedEx Ship Center, power plants, and truck traffic from the Hunts Point Food Distribution Center (Chen 2023)

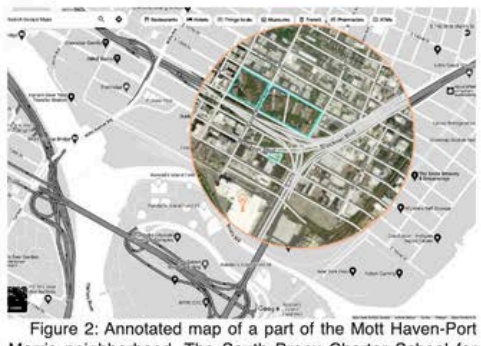


Figure 2: Annotated map of a part of the Mott Haven-Port Morris neighborhood. The South Bronx Charter School for International Culture and The Arts and the NYCHA Mill Brook Houses are enclosed in cyan (Chen 2023)

neighborhood waterfront a gray industrial zone owned by private entities. In 2020, FreshDirect delivered 100,000 grocery boxes to customers' doors each week (Business Insider 2020). About one thousand FreshDirect delivery trucks go in and out of the neighborhood daily across the two expressways and local roads, creating significantly increased truck and vehicle flow and upticks in air and noise pollution.

Data and traffic modeling from research institutions



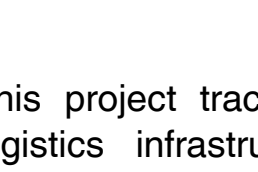


Figure 3: Data collection sites in Columbia University Mailman School of Public Health's study, including locations used for traffic counting, air quality measures, and noise measures around the FreshDirect warehouse. (Shearston et al. 2020).

	Site 3 Mean Count (SD)		Site 4 Mean Count (SD)	
	Pre-Warehouse Opening	Post-Warehouse Opening	Pre-Warehouse Opening	Post-Warehouse Opening
Trucks				
Time Window				
Midnight to 3 AM	66.2 (28.1)	63.4 (23.9)	5.8 (3.1)	5.4 (3.5)
3 to 6 AM	126.8 (59.2)	137.8 (59.4)	15.4 (6.6)	15.7 (4.3)
6 to 9 AM	249.2 (101.6)	263.1 (97.8)	34.3 (12.8)	34.3 (7.9)
9 AM to 12 noon	197.8 (102.1)	192.5 (59.4)	34.3 (7.8)	34.3 (7.9)
12 noon to 3 PM	268.3 (121.7)	303.4 (132.8)	17.0 (7.0)	19.3 (7.6)
3 to 6 PM	223.6 (96.9)	231.2 (102.3)	14.9 (6.2)	13.8 (5.7)
6 to 9 PM	104.4(43.4)	106.7 (45.1)	9.8 (4.3)	8.5 (4.0)
9 PM to Midnight	63.4(22.4)	65.1 (19.7)	7.6 (3.8)	10.4 (4.1)
Vehicles				
Time Window				
Midnight to 3 AM	492.7 (85.0)	470.6 (79.9)	55.0 (20.6)	54.6 (19.8)
3 to 6 AM	874.8 (282.1)	908.8 (266.4)	107.2 (36.5)	114.5 (34.8)
6 to 9 AM	1,294.4 (486.3)	1,354.6 (519.1)	215.5 (81.5)	224.5 (86.0)
9 AM to 12 noon	1,320.9 (363.1)	1,434.6 (390.7)	193.2 (71.9)	199.2 (76.8)
12 noon to 3 PM	1,445.7 (291.9)	1,556.8 (327.3)	202.7 (64.2)	224.6 (67.1)
3 to 6 PM	1,429.4 (238.7)	1,496.1 (244.8)	222.6 (87.0)	225.7 (79.9)
6 to 9 PM	1,076.1 (173.4)	1,139.6 (183.0)	136.6 (50.5)	154.1 (42.3)
9 PM to Midnight	728.4 (137.4)	710.7 (111.1)	93.6 (29.7)	93.3 (26.8)

Note: Pre-warehouse Opening time period = June 1, 2017 through September 30, 2018; Post-warehouse Opening time period = October 1, 2018 through May 5, 2019.

Figure 4: Table of descriptive statistics for measured truck and vehicle counts before and after the opening of FreshDirect in the South Bronx Mott Haven-Port Morris neighborhood. In both Site 3 and Site 4 as labeled in Figure 4, an increase in traffic flow is observed, leading to more congestion and noise and air pollution. (Shearston et al. 2020)

quantify these effects. Between June 2017 and May 2020, researchers at the Columbia University Mailman School of Public Health collected traffic, air quality, and noise measurements at eight sites near the warehouse, including four residential homes and a public housing complex (NYCHA Mill Brook Houses) as labeled in figure 4. They observed an overall traffic increase between ten and forty percent, with a more significant increase during overnight hours. The higher volume of traffic increases noise by 0.06 decibels A-weighted (dBA), leading to a lower quality of life due to worsened sleep. Increased diesel exhaust would increase black carbon air pollution by 0.003 μg/m3. Figure 5 shows air monitoring and noise monitoring sites and traffic counts as influenced by the opening of the FreshDirect warehouse (Shearston et al. 2020). Similar effects are propagated by truck traffic from the nearby Hunts Point Food Distribution Center. The Hunts Point Produce Market brings 15,000 diesel semi-trucks driving through the neighborhood to deliver food, causing higher traffic congestion (Gonzalez 2021).

To illustrate the lived experience of a typical resident in the neighborhood, I want to present two additional pieces of data. First is a photo collage with screenshots from Google Map Street View and a collection of photos on the web of the neighborhood. This collage emphasizes the epistemological experience of the community, of which there is a jarring juxtaposition between industrial pollution and the essential social fabric of the neighborhood. You see the South Bronx Charter School for International Culture and The Arts and the New York City Housing Authority (NYCHA) Mill Brook Houses, right next to the driving routes of diesel food trucks and gas-burning Peaker plants.

A student group in Columbia's Building Climate Justice: Co-Creative Coastal Resilience Planning class had a different idea when they made the ArcGIS StoryMap, named: "Mapping the Air: Illustrating Air Pollution Burden and Providing Recommendations for its Measurement in the South Bronx" (Mapping the Air 2023). After talking with South Bronx Unite, the students wanted to humanize the lived experiences of the South Bronx community by using portable personal air quality monitors called Atmotube PROs to map three typical routes in the neighborhood: a child in elementary school that suffers from asthma, a working adult, and an elderly person as shown in figure 6. The data they presented is concerning as some of the worst air quality measurements are found in proximity with school and parks and along routes that a child might take in their day-to-day life. Children are more likely to develop conditions such as acute respiratory infections, asthma, and decreased lung function caused by exposure to poor air quality.

PUBLIC HEALTH IMPACTS

To give a fuller picture of public health impacts due to truck exhaust and industrial siting, I will draw on data from the Climate and Environmental Justice Screen Tool (CEJST). CEJST is a federal level screening tool first released in November 2022. Issued by President Biden under Executive Order 14008 in January of 2021, the Council on Environmental Quality (CEQ) started to develop CEJST. The tool has an interactive map and uses datasets that are indicators of burdens in eight categories: climate change, energy, health, housing, legacy pollution, transportation, water and wastewater, and workforce development (About: Climate and Environmental Justice Screening Tool n.d.). According to CEJST, the two census tracts in the neighborhood are determined as disadvantaged because they are overburdened and underserved. In figures 8 and 9, transportation data—diesel particulate matter exposure, transportation barriers, traffic proximity and volume— and health data—asthma, diabetes, heart disease, and low life expectancy—are presented as the two census tracts near the highway junction.

In Mott Haven, diesel particulate matter exposure and asthma rates are in the ninety-seventh percentile compared to other neighborhoods. In 2020, the annual rate of asthma



Figure 5: Photo Collage and Illustration of Mott Haven-Port Morris neighborhood. The main pollution sites are integrated into the urban fabric of the neighborhood, with the South Bronx Charter School for International Culture and The Arts and the NYCHA Mill Brook Houses as major educational and residential communes nearby. Photos sourced from Google Maps Street View, edited with Adobe Photoshop and Procreate (Chen 2023)

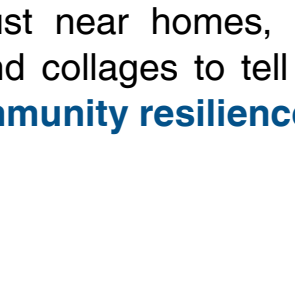


Figure 6: Design of daily routes for an elementary school student, a working adult, and an elderly person (Mapping the Air 2023).



Figure 7: Results of the $PM_{2.5}$ pollution level collected with Atmotube at locations among the routes (Mapping the Air 2023).



Figure 8: Tract information 1, annotated (CEJST).



Figure 9: Tract information 2, annotated (CEJST).

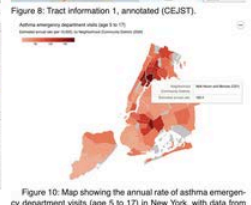


Figure 10: Map showing the annual rate of asthma emergency department visits (age 5 to 17) in New York, with data from the NYC Environment & Health Data Portal (Environment and Health Data Portal 2020).

emergency department visits for kids aged five to seventeen in the Mott Haven and Melrose area were significantly higher than in other community districts, as shown in figure 10. For kids and teenagers living and going to schools and parks in the area, air pollution can also be detrimental to their cognitive and physical development. Increasingly, research has suggested that air pollution does not just affect lungs but also influences childhood behavioral problems and even IQ. In "How Air Pollution Alters Brain Development: The Role of Neuroinflammation", published by the Translational Neuroscience journal, the study presents lines of converging evidence showing how several samples of children populations living in large cities around the world suffer to some degree neural, behavioral, and cognitive changes associated with air pollution exposure (Brockmeyer and D'Angiulli 2016). Another study, published in Environmental Health Perspectives, found that children whose mothers experienced higher nitrogen dioxide (NO_2) exposure during pregnancy, particularly in the first and second trimesters, were more likely to have behavioral problems. Higher exposures to small-particle air pollution ($PM_{2.5}$) when children were two to four years old was associated with poorer child behavioral functioning and cognitive performance (Science Daily 2022). Poorer cognitive performance could lead to lower academic achievement as children struggle with concentrating and learning, with long-term impacts on memory and attention.

HISTORY OF REDLINING AND COMMUNITY RESISTANCE

The history of environmental pollution in Mott Haven-Port Morris has been linked with discriminatory zoning. The neighborhood was marked "RED" in the Redlining maps created by the Home Owners' Loan Corporation (HOLC) as part of its City Survey Program in the late 1930s. As shown in figure 11, the Bronx neighborhood is classified to have the "detrimental influence" of "Industrial encroachment. Elevated R.R. Heavy Traffic. High Congested" and was predicted to have a "downfast" trend of desirability in the next ten to fifteen years. Hence, poor lending opportunities led to disinvestment, causing the area's housing segregation and economic desperation. In 1939, the six-lane Major Deegan Expressway was constructed, and in 1962, the six-lane Bruckner Expressway— both increasing diesel traffic

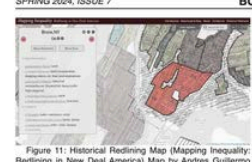


Figure 11: Historical Redlining Map (Mapping Inequality: Redlining in New Deal America) Map by Andres Guillermo Blanco, 2010. The Formal Determinants of Informal Settlements in Bogota, Colombia. Ithaca, NY: Cornell University.

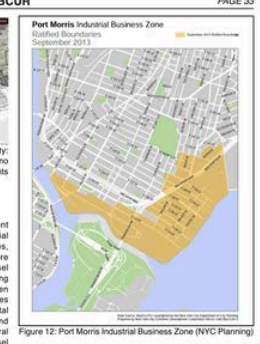


Figure 12: Port Morris Industrial Business Zone (NYC Planning)

(Columbia Public Health 2022).

In 2013, the New York City Economic Development Corporation (NYCEDC) assigned Port Morris as an Industrial Business Zone. Besides three highways and five bridges, New York State and City encouraged FreshDirect with more than $150 million in subsidies in 2014 to relocate their diesel trucking warehouse to the South Bronx waterfront, bringing more industrial contamination to Mott Haven-Port Morris when the area was already heavily polluted by toxic Superfund sites and diesel truck emissions. In the "South Bronx Environmental Health and Policy Study" twenty years ago, the EPA found that particulate matter in the area frequently exceeded federal limits and that asthma levels directly correlated with diesel truck emissions from the area's highway and industrial facility saturation. Despite FreshDirect's promise of converting to electric trucks, it seems to be more of a public-facing statement for its brand identity than a commitment to electrifying its fleet to reduce environmental and health harms. After all, FreshDirect wants visibility on the "freshness" of its groceries to their customers than the air pollution, noise pollution, and negative health effects caused by its grocery delivering trucks.

Local journalism reveals the empty promise of FreshDirect. Welcome2TheBronx is a news site founded in September 2009 by Editor and Senior Writer Ed Garcia Conde (Welcome2TheBronx n.d.). With a local ambition to fill the digital void in mainstream media and combat decades-old stereotypes about the borough, the news site publishes detailed coverage and reporting on a wide range of topics in the Bronx, from arts and culture, preservation issues, to affordable housing, gentrification, and feel good stories about Bronx residents. Since 2009, it has grown to become a credible and meaningful information source, acknowledged by the New York City Historic District Council; the Avery Library Historic Preservation and Urban Planning web archive; and cited in over five hundred articles across major mainstream media news sites like The New York Times, The New York Daily News, and The New York Post. In 2012, due to Superstorm Sandy, over one hundred FreshDirect trucks were damaged and dumped on the Mott Haven-Port Morris waterfront. Conde writes that during that time, FreshDirect had a chance to purchase one hundred electric trucks to comply with their promise and they did not. In 2014, due to local concerned citizens taking photographs of new FreshDirect trucks, Conde was able to investigate and report on fifteen brand new diesel trucks being purchased by FreshDirect, most of them having "certified clean idling" stickers meaning that the trucks do not pollute when they are stationary, but still spew out diesel fumes when they are being driven around. Conde calls out FreshDirect's lies as well as politicians and organizations that receive ads and sponsorships from FreshDirect which support its continuous "life-threatening" operation (Conde 2014).

Moreover, the history of redlining and disproportionate health burdens have spurred a strong momentum of community organizing. Since its founding in 2012, local organization South Bronx Unite has been fighting the FreshDirect relocation. They testified at public hearings, organized demonstrations and marches, attracted media attention, wrote op-eds, reached out to local legislators, and filed a lawsuit against the city and the State in 2013 because of their failure to conduct an adequate environmental review of the site—choosing to use a 21 year old environmental impact statement—in the attempt to boycott the plan (South Bronx Unite n.d.). Although these efforts failed to stop FreshDirect, it ignited community resistance to discriminatory zoning and fueled the fight for cleaner air, greener spaces, and better jobs in Mott Haven-Port Morris today. WE ACT for Environmental Justice also



Figure 13: Collage of local resistance efforts to the relocation of FreshDirect to the South Bronx (Chen 2023)



Figure 14: South Bronx Unite Toxic Tour - WE ACT for Environmental Justice (South Bronx Unite Toxic Tour n.d.)

co-organizes South Bronx Unite Toxic Tour to reveal the sites of environmental racism and injustices in the South Bronx, one of the routes shown in figure 14.

These acts of local journalism to create community-owned data and community organizing send a clear message: South Bronx will not allow itself to remain a silent sacrifice zone. Environmental reparations involve putting stricter regulations on polluting corporations; bringing green and future-proof jobs and education programs to the neighborhood; and improving public infrastructure and public health. Community, academic, and municipal institutions need to work together in creating better futures of living and working in the community.

In February 2024, 93-year-old Ruth Gottesman, made a historic $1 billion donation to the Bronx's Albert Einstein College of Medicine to provide free tuition for all medical students. According to a press release from Einstein College of Medicine, "This donation radically revolutionizes our ability to continue attracting students who are committed to our mission, not just those who can afford it. Additionally, it will free up and lift our students, enabling them to pursue projects and ideas that might otherwise be prohibitive," said Dr. Yaron Tomer, the Marilyn and Stanley Katz Dean at Albert Einstein College of Medicine (Conde 2024). Students who aspire to be doctors can attend medical school in their own borough for free. This will hopefully have a transformative impact on health care outcomes in the neighborhood and repair the historical legacies of environmental racism and injustice.

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3.4

SPATIAL REGRESSION: RETHINKING PROXIMITY AND RISK IN SAN FRANCISCO

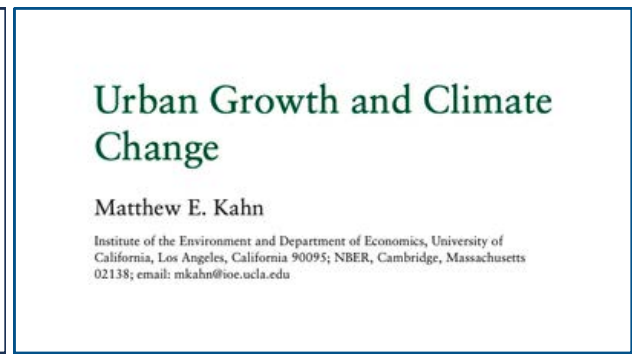
Project Type: Academic; Collaborative
Course: Advanced GIS
Supervisor: Prof. Leah Meisterlin
Date: Feb. – May 2025

This study looks at **how economic vulnerability and property values are distributed in relation to flood risk in San Francisco**, using block-group-level data and several proximity measures, including elevation, horizontal distance, and combined 3D distance. The results provide little support for the idea that lower-income areas are closer to flood zones, and rent burden does not show a clear spatial pattern. Instead, land and property values often go against this idea: the most valuable areas are usually closest to flood risk, especially along the waterfront.

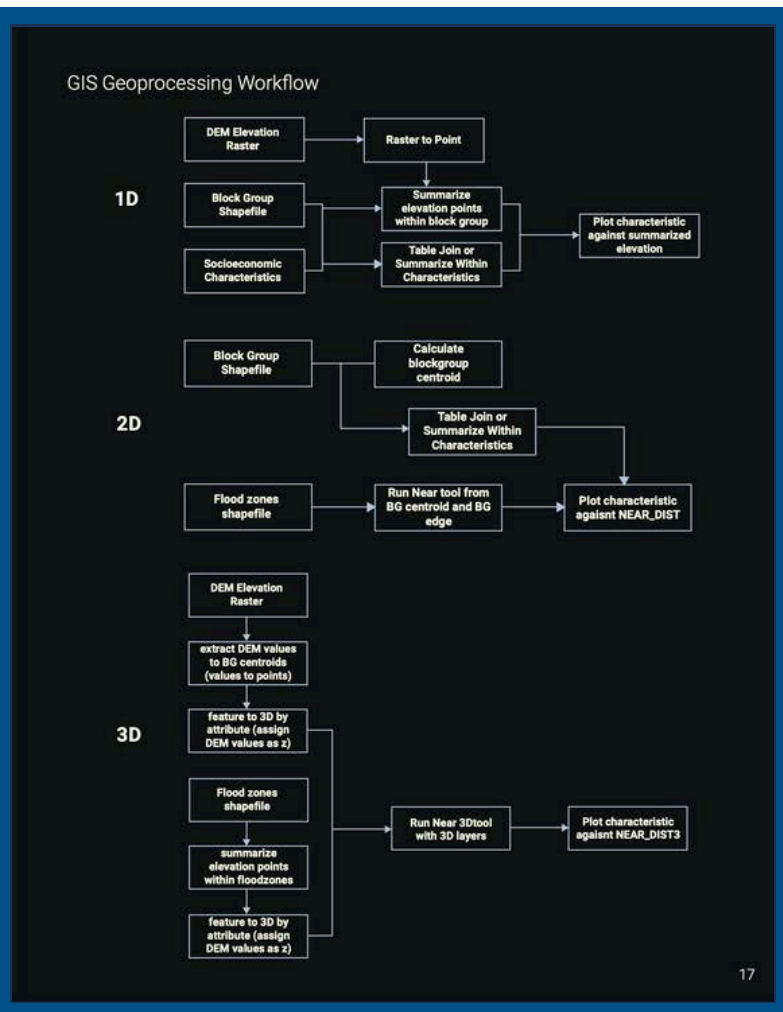
These patterns raise concerns about the **moral hazard of investing in resilience**, as coastal protection might encourage more high-value development in risky areas. San Francisco's small, island-like geography also creates **challenges for research methods**, which suggests the need for future studies using better spatial metrics and wider regional comparisons.

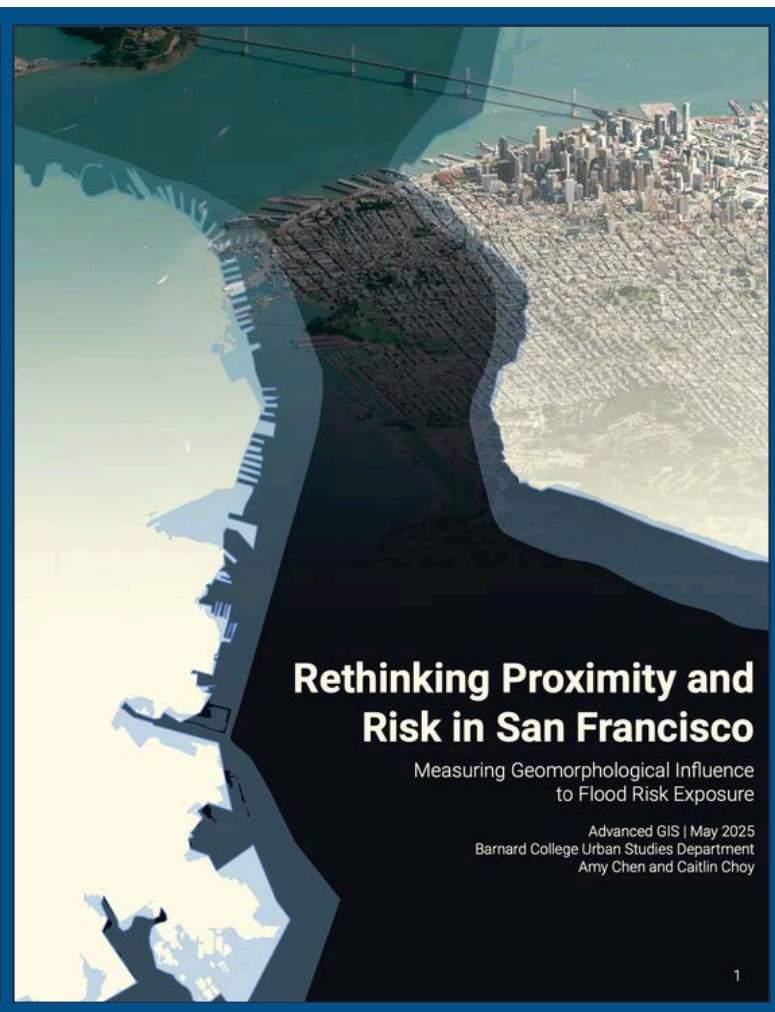
Role: Developed the research concept; conducted ArcGIS analysis and spatial plotting; led layout design and writing.

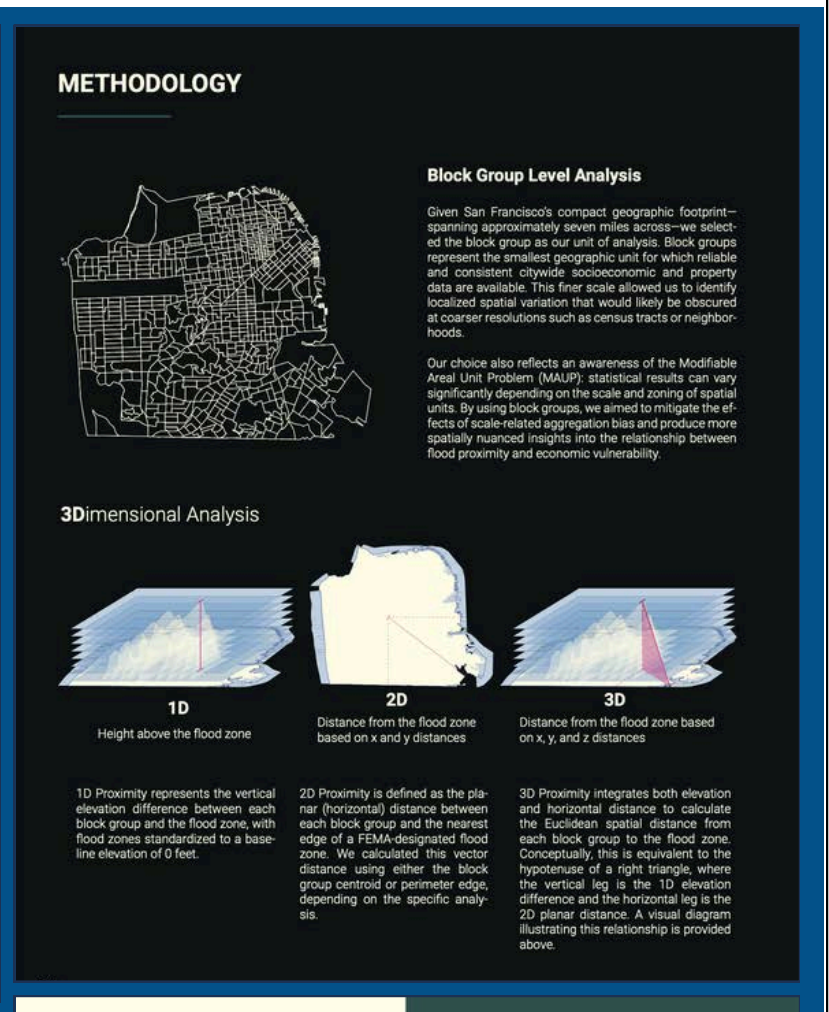




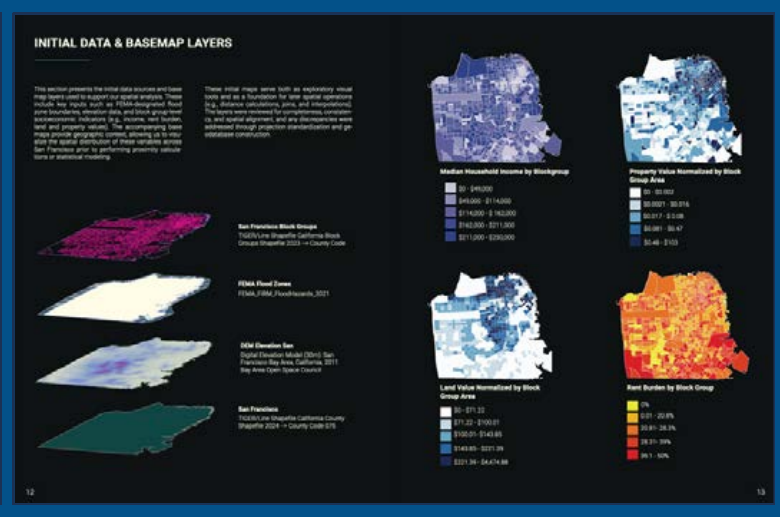
Inspiration for the project (an economics paper by Matthew Kahn)

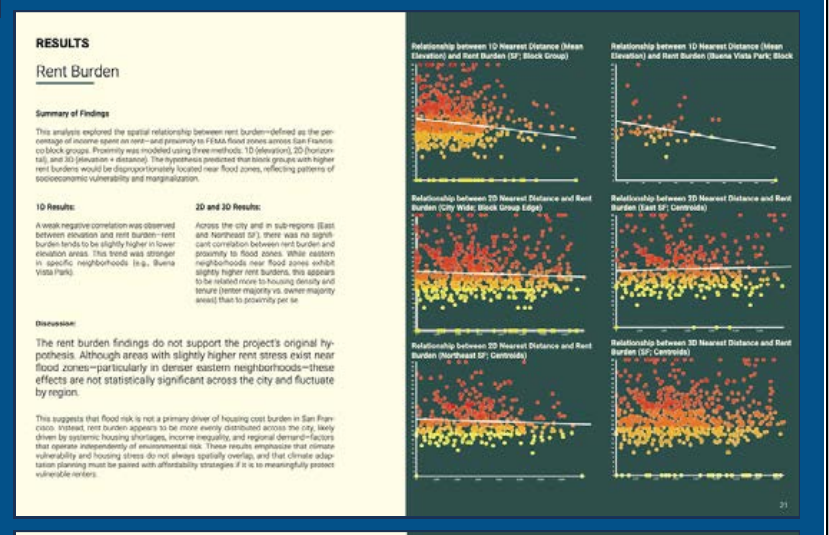


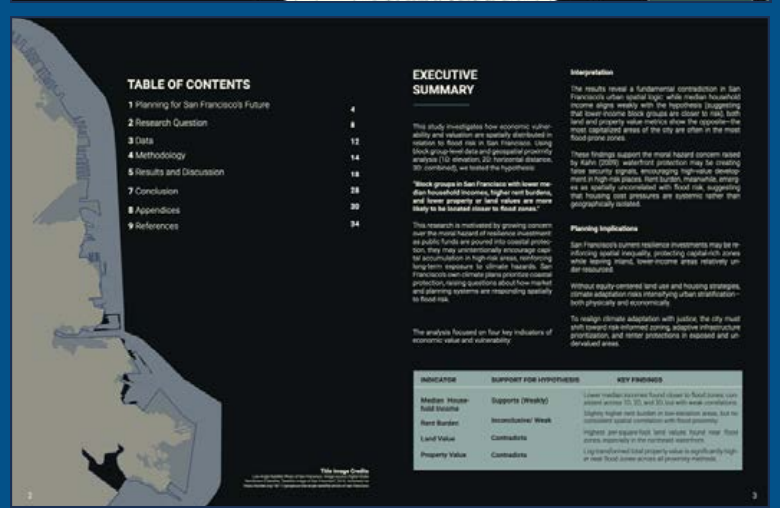


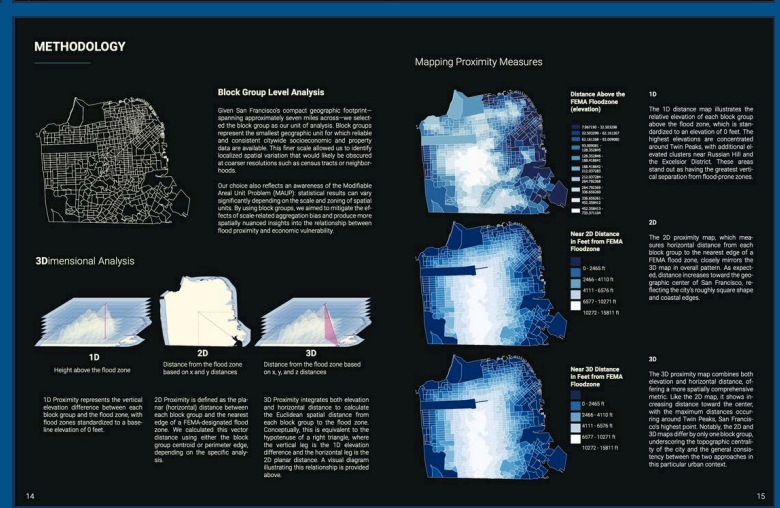


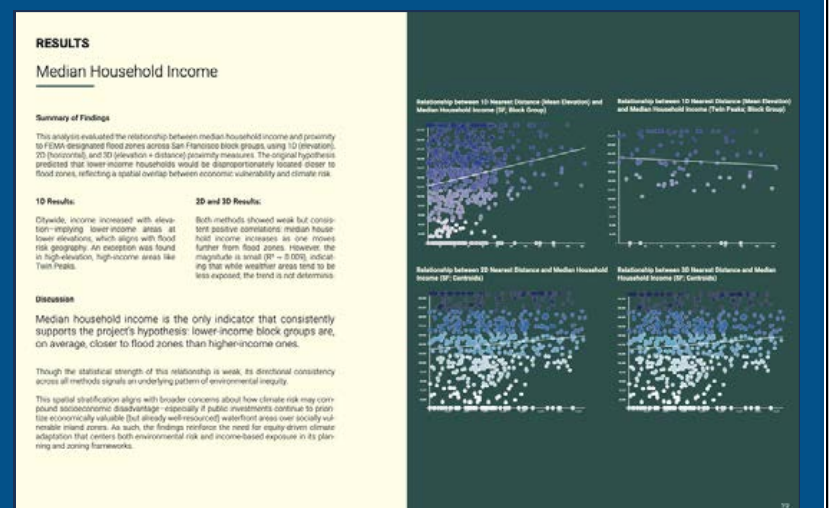












3.5

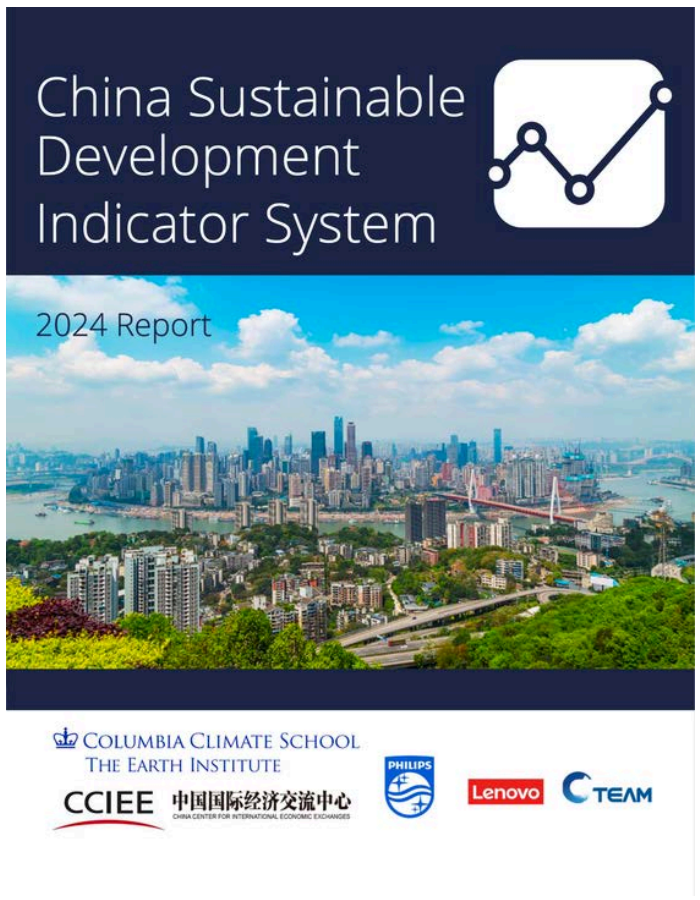
POLICY CASE STUDIES ON SUSTAINABLE CITIES

Project Type: Office-related
Sustainability Policy and Management Research Program, Columbia Climate School, New York
Date: 2023 – 2025

As **Curriculum Co-Developer for a Tencent-funded High School Sustainability Education Project,** I authored a **200-page section on waste management systems in the Sustainable Cities chapter**. The work involved researching and writing about waste, creating case studies, and designing applied exercises. I integrated many interdisciplinary approaches that make the study of waste more interesting.

I also served as a **Research Assistant on the China Sustainable Development Indicator System, contributing to national-level sustainability reporting linked to COP29 and COP30**. My work included policy synthesis and translation for the 2024 national indicator report, as well as the development of comparative policy case studies and data profiles on Lima and Copenhagen for the 2025 global sustainable development reports. These findings were presented at COP in Baku, supporting international dialogue on sustainability metrics and policy alignment.

China Sustainable Development Indicator System





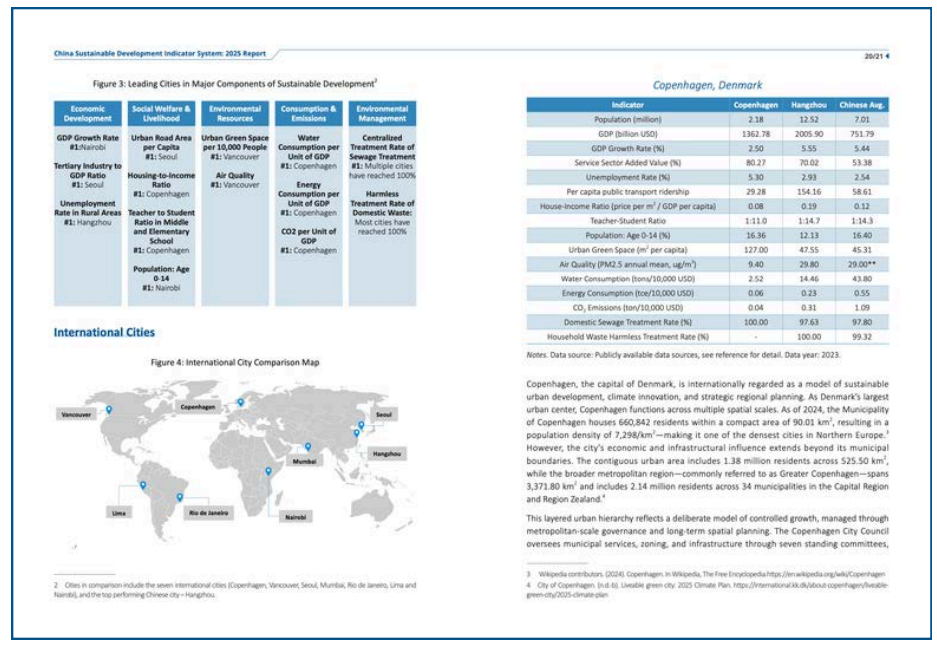

China Sustainable Development Indicator System: 2025 Report 20/21

Figure 3: Leading Cities in Major Components of Sustainable Development[2]

Economic Development	Social Welfare & Livelihood	Environmental Resources	Consumption & Emissions	Environmental Management
GDP Growth Rate #1: Nairobi Tertiary Industry to GDP Ratio #1: Seoul Unemployment Rate in Rural Areas #1: Hangzhou	Urban Road Area per Capita #1: Seoul Housing-to-Income Ratio #1: Copenhagen Teacher to Student Ratio in Middle and Elementary School #1: Copenhagen Population: Age 0-14 #1: Nairobi	Urban Green Space per 10,000 People #1: Vancouver Air Quality #1: Vancouver	Water Consumption per Unit of GDP #1: Copenhagen Energy Consumption per Unit of GDP #1: Copenhagen CO2 per Unit of GDP #1: Copenhagen	Centralized Treatment Rate of Sewage Treatment #1: Multiple cities have reached 100% Harmless Treatment Rate of Domestic Waste: Most cities have reached 100%

International Cities

Figure 4: International City Comparison Map

2 Cities in comparison include the seven international cities (Copenhagen, Vancouver, Seoul, Mumbai, Rio de Janeiro, Lima and Nairobi), and the top performing Chinese city – Hangzhou.

Copenhagen, Denmark

Indicator	Copenhagen	Hangzhou	Chinese Avg.
Population (million)	2.18	12.52	7.01
GDP (billion USD)	1362.78	2005.90	751.79
GDP Growth Rate (%)	2.50	5.55	5.44
Service Sector Added Value (%)	80.27	70.02	53.38
Unemployment Rate (%)	5.30	2.93	2.54
Per capita public transport ridership	29.28	154.16	58.61
House-Income Ratio (price per m^2 / GDP per capita)	0.08	0.19	0.12
Teacher-Student Ratio	1:11.0	1:14.7	1:14.3
Population: Age 0-14 (%)	16.36	12.13	16.40
Urban Green Space (m^2 per capita)	127.00	47.55	45.31
Air Quality (PM2.5 annual mean, ug/m^3)	9.40	29.80	29.00**
Water Consumption (tons/10,000 USD)	2.52	14.46	43.80
Energy Consumption (tce/10,000 USD)	0.06	0.23	0.55
CO_2 Emissions (ton/10,000 USD)	0.04	0.31	1.09
Domestic Sewage Treatment Rate (%)	100.00	97.63	97.80
Household Waste Harmless Treatment Rate (%)	-	100.00	99.32

Notes: Data source: Publicly available data sources, see reference for detail. Data year: 2023.

Copenhagen, the capital of Denmark, is internationally regarded as a model of sustainable urban development, climate innovation, and strategic regional planning. As Denmark's largest urban center, Copenhagen functions across multiple spatial scales. As of 2024, the Municipality of Copenhagen houses 660,842 residents within a compact area of 90.01 km^2, resulting in a population density of 7,298/km^2—making it one of the densest cities in Northern Europe.[3] However, the city's economic and infrastructural influence extends beyond its municipal boundaries. The contiguous urban area includes 1.38 million residents across 525.50 km^2, while the broader metropolitan region—commonly referred to as Greater Copenhagen—spans 3,371.80 km^2 and includes 2.14 million residents across 34 municipalities in the Capital Region and Region Zealand.[4]

This layered urban hierarchy reflects a deliberate model of controlled growth, managed through metropolitan-scale governance and long-term spatial planning. The Copenhagen City Council oversees municipal services, zoning, and infrastructure through seven standing committees,

3 Wikipedia contributors. (2024). Copenhagen. In Wikipedia, The Free Encyclopedia https://en.wikipedia.org/wiki/Copenhagen
4 City of Copenhagen. (n.d-b). Liveable green city: 2025 Climate Plan. https://international.kk.dk/about-copenhagen/liveable-green-city/2025-climate-plan

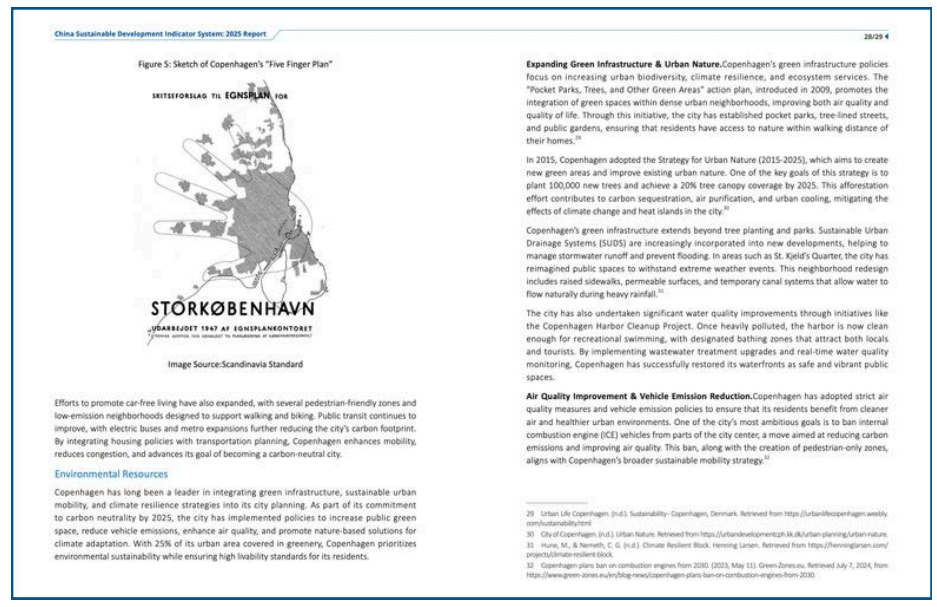

China Sustainable Development Indicator System: 2025 Report 28/29

Figure 5: Sketch of Copenhagen's "Five Finger Plan"

Image Source:Scandinavia Standard

Efforts to promote car-free living have also expanded, with several pedestrian-friendly zones and low-emission neighborhoods designed to support walking and biking. Public transit continues to improve, with electric buses and metro expansions further reducing the city's carbon footprint. By integrating housing policies with transportation planning, Copenhagen enhances mobility, reduces congestion, and advances its goal of becoming a carbon-neutral city.

Environmental Resources

Copenhagen has long been a leader in integrating green infrastructure, sustainable urban mobility, and climate resilience strategies into its city planning. As part of its commitment to carbon neutrality by 2025, the city has implemented policies to increase public green space, reduce vehicle emissions, enhance air quality, and promote nature-based solutions for climate adaptation. With 25% of its urban area covered in greenery, Copenhagen prioritizes environmental sustainability while ensuring high livability standards for its residents.

Expanding Green Infrastructure & Urban Nature. Copenhagen's green infrastructure policies focus on increasing urban biodiversity, climate resilience, and ecosystem services. The "Pocket Parks, Trees, and Other Green Areas" action plan, introduced in 2009, promotes the integration of green spaces within dense urban neighborhoods, improving both air quality and quality of life. Through this initiative, the city has established pocket parks, tree-lined streets, and public gardens, ensuring that residents have access to nature within walking distance of their homes.[29]

In 2015, Copenhagen adopted the Strategy for Urban Nature (2015-2025), which aims to create new green areas and improve existing urban nature. One of the key goals of this strategy is to plant 100,000 new trees and achieve a 20% tree canopy coverage by 2025. This afforestation effort contributes to carbon sequestration, air purification, and urban cooling, mitigating the effects of climate change and heat islands in the city.[30]

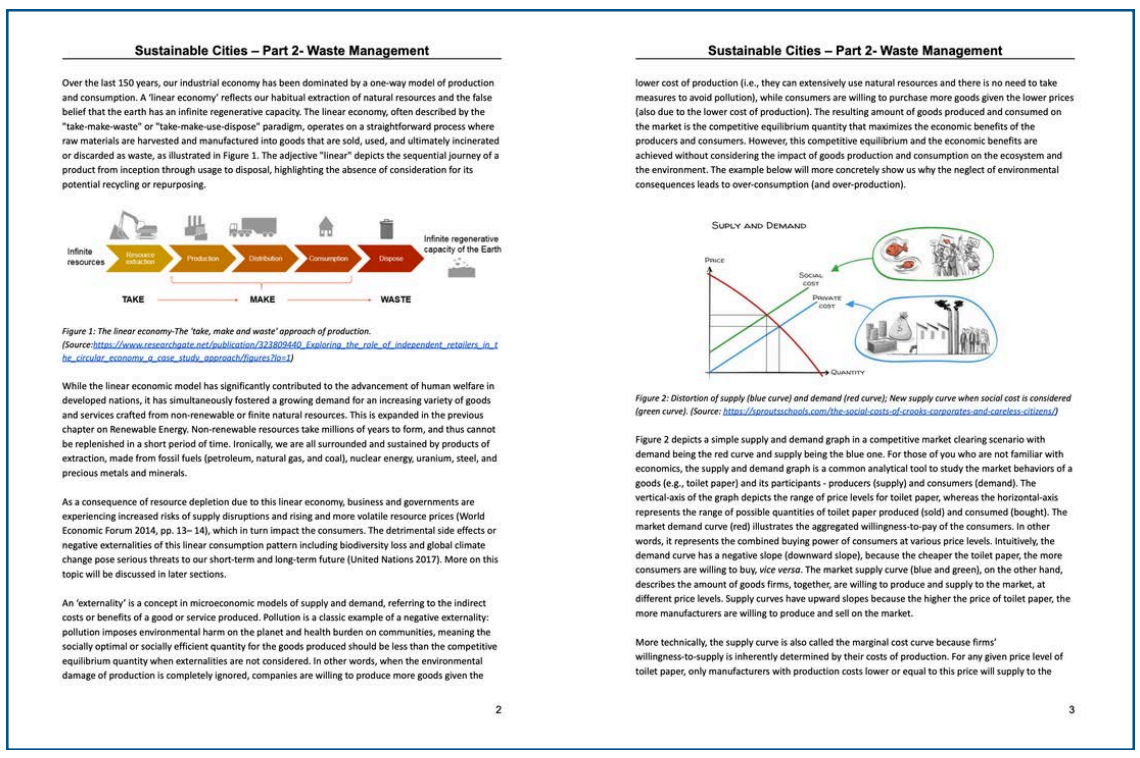
Copenhagen's green infrastructure extends beyond tree planting and parks. Sustainable Urban Drainage Systems (SUDS) are increasingly incorporated into new developments, helping to manage stormwater runoff and prevent flooding. In areas such as St. Kjeld's Quarter, the city has reimagined public spaces to withstand extreme weather events. This neighborhood redesign includes raised sidewalks, permeable surfaces, and temporary canal systems that allow water to flow naturally during heavy rainfall.[31]

The city has also undertaken significant water quality improvements through initiatives like the Copenhagen Harbor Cleanup Project. Once heavily polluted, the harbor is now clean enough for recreational swimming, with designated bathing zones that attract both locals and tourists. By implementing wastewater treatment upgrades and real-time water quality monitoring, Copenhagen has successfully restored its waterfronts as safe and vibrant public spaces.

Air Quality Improvement & Vehicle Emission Reduction. Copenhagen has adopted strict air quality measures and vehicle emission policies to ensure that its residents benefit from cleaner air and healthier urban environments. One of the city's most ambitious goals is to ban internal combustion engine (ICE) vehicles from parts of the city center, a move aimed at reducing carbon emissions and improving air quality. This ban, along with the creation of pedestrian-only zones, aligns with Copenhagen's broader sustainable mobility strategy.[32]

29 Urban Life Copenhagen. (n.d.). Sustainability- Copenhagen, Denmark. Retrieved from https://urbanlifecopenhagen.weebly.com/sustainability.html
30 City of Copenhagen. (n.d.). Urban Nature. Retrieved from https://urbandevelopmentcph.kk.dk/urban-planning/urban-nature.
31 Hune, M., & Nemeth, C. G. (n.d.). Climate Resilient Block. Henning Larsen. Retrieved from https://henninglarsen.com/projects/climate-resilient-block.
32 Copenhagen plans ban on combustion engines from 2030. (2023, May 11). Green-Zones.eu. Retrieved July 7, 2024, from https://www.green-zones.eu/en/blog-news/copenhagen-plans-ban-on-combustion-engines-from-2030.

Waste Management Section, Sustainable


Sustainable Cities – Part 2- Waste Management

Over the last 150 years, our industrial economy has been dominated by a one-way model of production and consumption. A 'linear economy' reflects our habitual extraction of natural resources and the false belief that the earth has an infinite regenerative capacity. The linear economy, often described by the "take-make-waste" or "take-make-use-dispose" paradigm, operates on a straightforward process where raw materials are harvested and manufactured into goods that are sold, used, and ultimately incinerated or discarded as waste, as illustrated in Figure 1. The adjective "linear" depicts the sequential journey of a product from inception through usage to disposal, highlighting the absence of consideration for its potential recycling or repurposing.

Figure 1: The linear economy-The 'take, make and waste' approach of production. (Source:https://www.researchgate.net/publication/323809440_Exploring_the_role_of_independent_retailers_in_the_circular_economy_a_case_study_approach/figures?lo=1)

While the linear economic model has significantly contributed to the advancement of human welfare in developed nations, it has simultaneously fostered a growing demand for an increasing variety of goods and services crafted from non-renewable or finite natural resources. This is expanded in the previous chapter on Renewable Energy. Non-renewable resources take millions of years to form, and thus cannot be replenished in a short period of time. Ironically, we are all surrounded and sustained by products of extraction, made from fossil fuels (petroleum, natural gas, and coal), nuclear energy, uranium, steel, and precious metals and minerals.

As a consequence of resource depletion due to this linear economy, business and governments are experiencing increased risks of supply disruptions and rising and more volatile resource prices (World Economic Forum 2014, pp. 13– 14), which in turn impact the consumers. The detrimental side effects or negative externalities of this linear consumption pattern including biodiversity loss and global climate change pose serious threats to our short-term and long-term future (United Nations 2017). More on this topic will be discussed in later sections.

An 'externality' is a concept in microeconomic models of supply and demand, referring to the indirect costs or benefits of a good or service produced. Pollution is a classic example of a negative externality: pollution imposes environmental harm on the planet and health burden on communities, meaning the socially optimal or socially efficient quantity for the goods produced should be less than the competitive equilibrium quantity when externalities are not considered. In other words, when the environmental damage of production is completely ignored, companies are willing to produce more goods given the

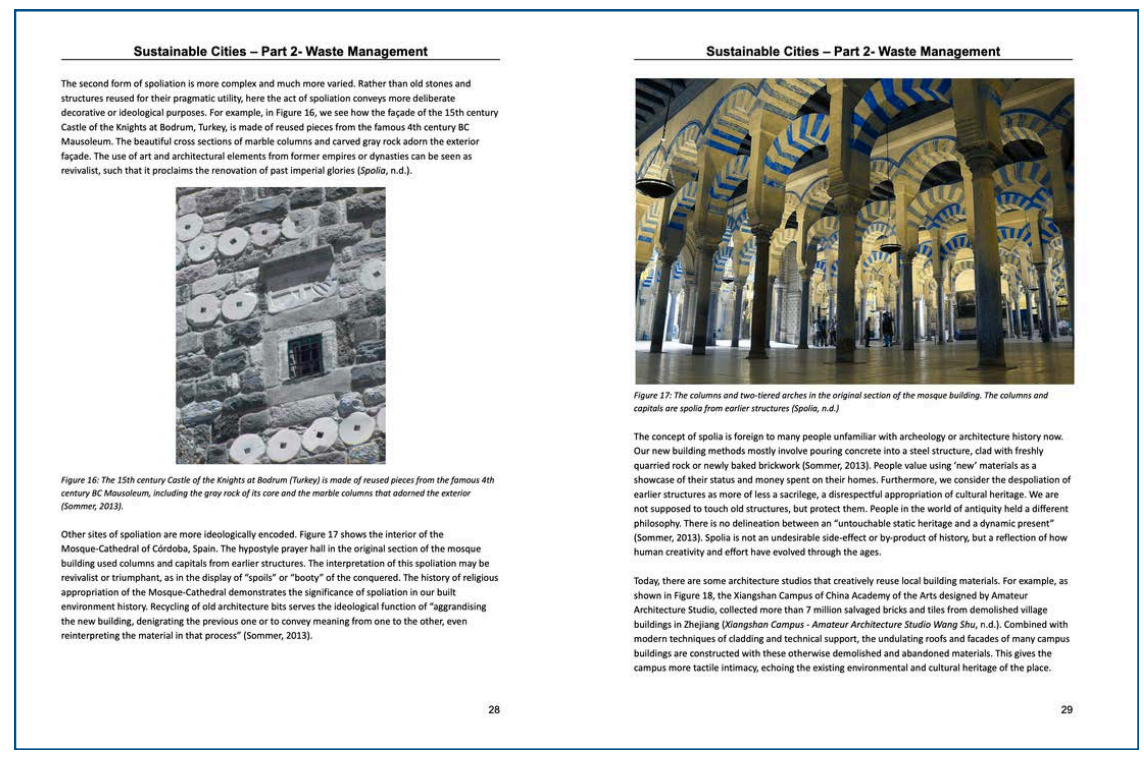
Sustainable Cities – Part 2- Waste Management

lower cost of production (i.e., they can extensively use natural resources and there is no need to take measures to avoid pollution), while consumers are willing to purchase more goods given the lower prices (also due to the lower cost of production). The resulting amount of goods produced and consumed on the market is the competitive equilibrium quantity that maximizes the economic benefits of the producers and consumers. However, this competitive equilibrium and the economic benefits are achieved without considering the impact of goods production and consumption on the ecosystem and the environment. The example below will more concretely show us why the neglect of environmental consequences leads to over-consumption (and over-production).

Figure 2: Distortion of supply (blue curve) and demand (red curve); New supply curve when social cost is considered (green curve). (Source: https://sproutsschools.com/the-social-costs-of-crooks-corporates-and-careless-citizens/)

Figure 2 depicts a simple supply and demand graph in a competitive market clearing scenario with demand being the red curve and supply being the blue one. For those of you who are not familiar with economics, the supply and demand graph is a common analytical tool to study the market behaviors of a goods (e.g., toilet paper) and its participants - producers (supply) and consumers (demand). The vertical-axis of the graph depicts the range of price levels for toilet paper, whereas the horizontal-axis represents the range of possible quantities of toilet paper produced (sold) and consumed (bought). The market demand curve (red) illustrates the aggregated willingness-to-pay of the consumers. In other words, it represents the combined buying power of consumers at various price levels. Intuitively, the demand curve has a negative slope (downward slope), because the cheaper the toilet paper, the more consumers are willing to buy, *vice versa*. The market supply curve (blue and green), on the other hand, describes the amount of goods firms, together, are willing to produce and supply to the market, at different price levels. Supply curves have upward slopes because the higher the price of toilet paper, the more manufacturers are willing to produce and sell on the market.

More technically, the supply curve is also called the marginal cost curve because firms' willingness-to-supply is inherently determined by their costs of production. For any given price level of toilet paper, only manufacturers with production costs lower or equal to this price will supply to the


Sustainable Cities – Part 2- Waste Management

The second form of spoliation is more complex and much more varied. Rather than old stones and structures reused for their pragmatic utility, here the act of spoliation conveys more deliberate decorative or ideological purposes. For example, in Figure 16, we see how the façade of the 15th century Castle of the Knights at Bodrum, Turkey, is made of reused pieces from the famous 4th century BC Mausoleum. The beautiful cross sections of marble columns and carved gray rock adorn the exterior façade. The use of art and architectural elements from former empires or dynasties can be seen as revivalist, such that it proclaims the renovation of past imperial glories (*Spolia*, n.d.).

Figure 16: The 15th century Castle of the Knights at Bodrum (Turkey) is made of reused pieces from the famous 4th century BC Mausoleum, including the gray rock of its core and the marble columns that adorned the exterior (Sommer, 2013).

Other sites of spoliation are more ideologically encoded. Figure 17 shows the interior of the Mosque-Cathedral of Córdoba, Spain. The hypostyle prayer hall in the original section of the mosque building used columns and capitals from earlier structures. The interpretation of this spoliation may be revivalist or triumphant, as in the display of "spoils" or "booty" of the conquered. The history of religious appropriation of the Mosque-Cathedral demonstrates the significance of spoliation in our built environment history. Recycling of old architecture bits serves the ideological function of "aggrandising the new building, denigrating the previous one or to convey meaning from one to the other, even reinterpreting the material in that process" (Sommer, 2013).

Sustainable Cities – Part 2- Waste Management

Figure 17: The columns and two-tiered arches in the original section of the mosque building. The columns and capitals are spolia from earlier structures (Spolia, n.d.)

The concept of spolia is foreign to many people unfamiliar with archeology or architecture history now. Our new building methods mostly involve pouring concrete into a steel structure, clad with freshly quarried rock or newly baked brickwork (Sommer, 2013). People value using 'new' materials as a showcase of their status and money spent on their homes. Furthermore, we consider the despoliation of earlier structures as more of less a sacrilege, a disrespectful appropriation of cultural heritage. We are not supposed to touch old structures, but protect them. People in the world of antiquity held a different philosophy. There is no delineation between an "untouchable static heritage and a dynamic present" (Sommer, 2013). Spolia is not an undesirable side-effect or by-product of history, but a reflection of how human creativity and effort have evolved through the ages.

Today, there are some architecture studios that creatively reuse local building materials. For example, as shown in Figure 18, the Xiangshan Campus of China Academy of the Arts designed by Amateur Architecture Studio, collected more than 7 million salvaged bricks and tiles from demolished village buildings in Zhejiang (*Xiangshan Campus - Amateur Architecture Studio Wang Shu*, n.d.). Combined with modern techniques of cladding and technical support, the undulating roofs and facades of many campus buildings are constructed with these otherwise demolished and abandoned materials. This gives the campus more tactile intimacy, echoing the existing environmental and cultural heritage of the place.

3.6

HOUSING POLICY ALIGNMENT ANALYTICS USING SENTIMENT METHODS

Methods: NYT Archive API ingestion, keyword and geographic filtering, VADER with custom valence dictionaries and period-conditional scoring, word-count-weighted aggregation with rolling smoothing, rule-based frame classification with κ validation, Bai–Perron structural break estimation with HAC errors, LaTeX.

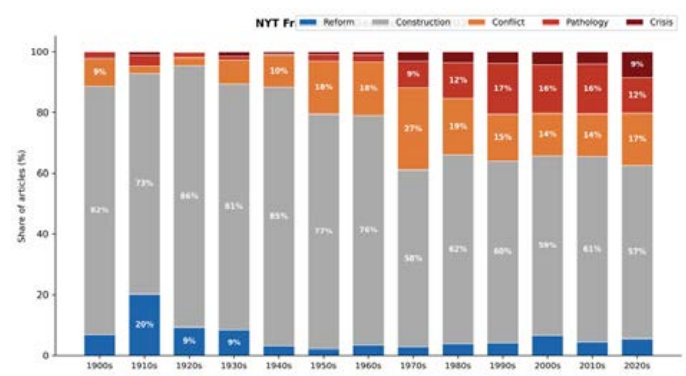
Role: Sole author: research question, corpus construction, lexicon design, estimation, mapping, and write-up.

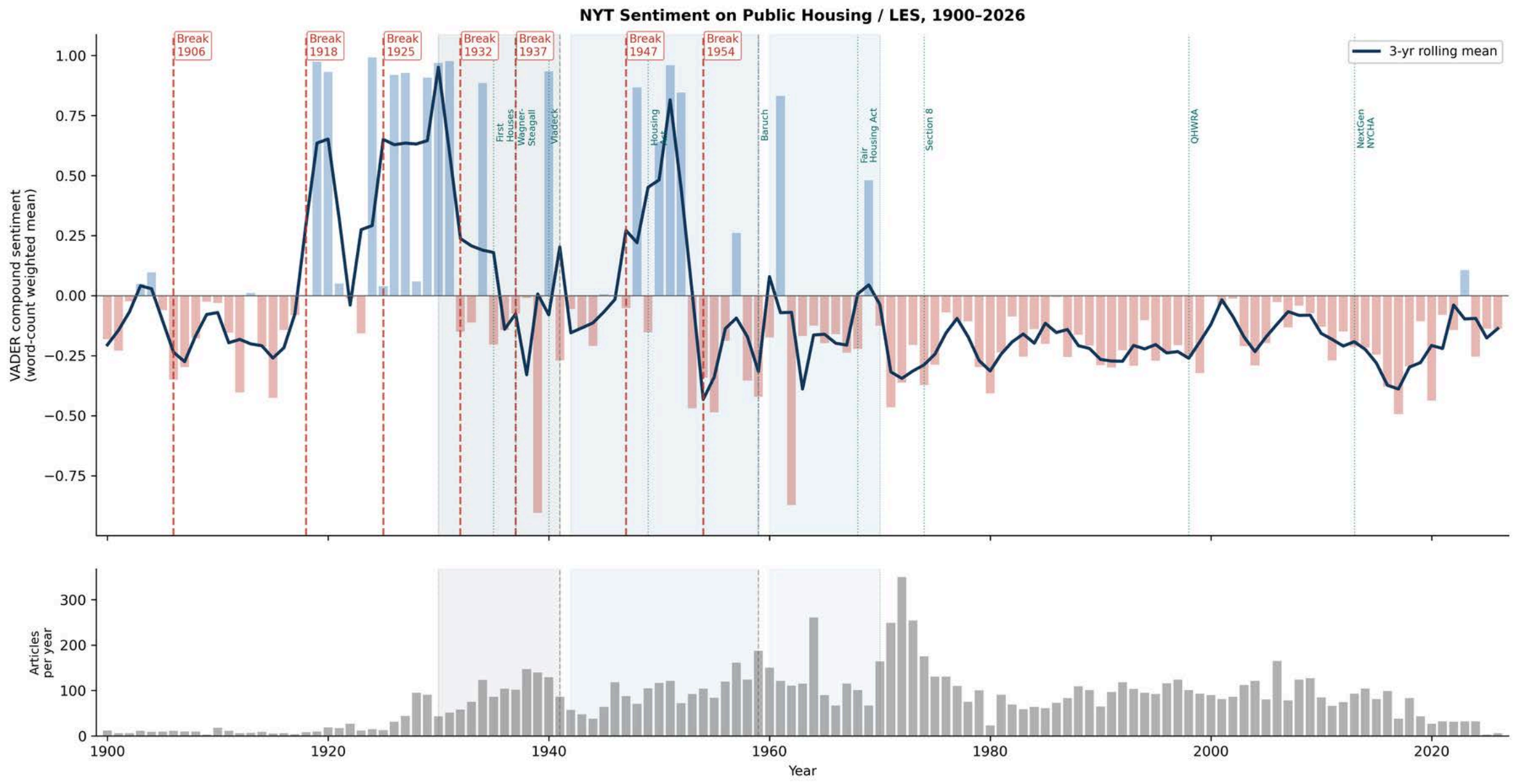
I assembled 10,118 New York Times articles from January 1900 to April 2026 via the Archive API, filtered to news, editorial, and review material matching both a housing keyword and a Manhattan or Lower East Side geographic referent. Articles are scored with VADER under three domain augmentations: added valence for housing-reform vocabulary, added valence for housing-pathology vocabulary, and— the augmentation the project turns on— conditional scoring for semantic drift, with project neutral before 1950 and negative after, and clearance positive before 1955 and negative thereafter.

The result of the sentiment analysis is a fresh look into how public's perception on the public housing program in New York City has changed, based on the subset of data in Lower East Side. This could justify innovations in how NYCHA addresses its maintenance deficit, through programs like the Permanent Affordability Commitment Together (PACT).









METHODOLOGY APPENDIX: SPATIAL REPRESENTATION & FIELD OBSERVATION

M1

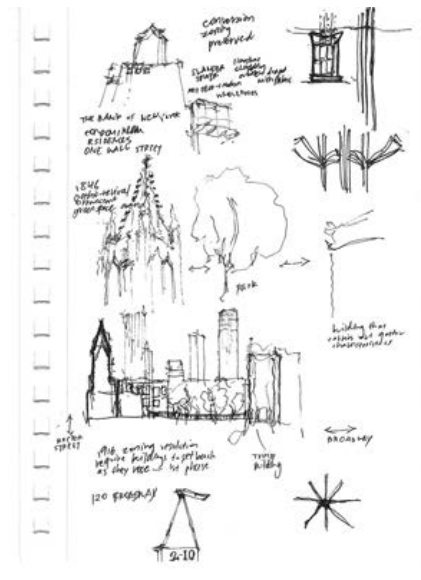
OBSERVATIONAL DRAWING

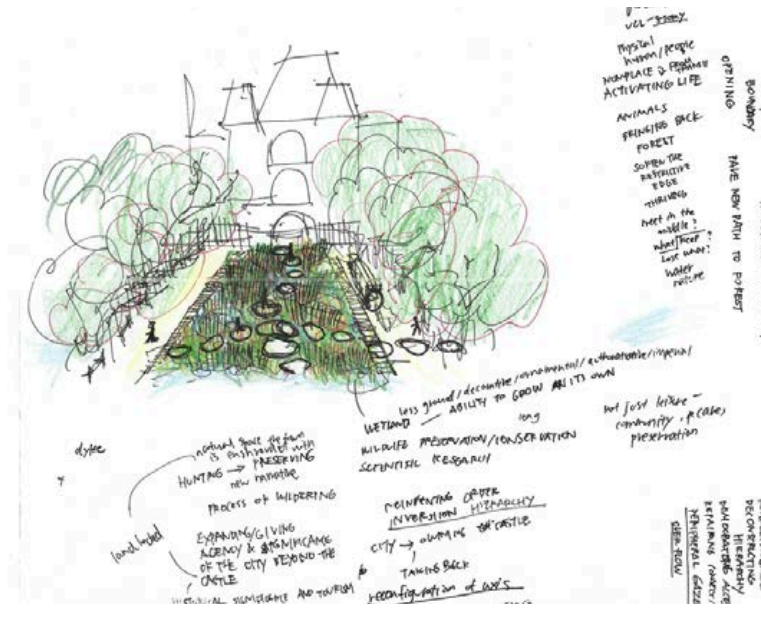
I have always loved **drawing from life**. It has taught me how to capture the **essence of a place and the color of a memory** that jumps out vividly from a page, even when I open the sketchbook a few years later.

Project Type: Personal
Site: Château de Fontainebleau, France; Notre Dame du Haut (Ronchamp), France; Gilder Center, American Museum of Natural History, New York; somewhere in Central Park
Date: 2023–2025









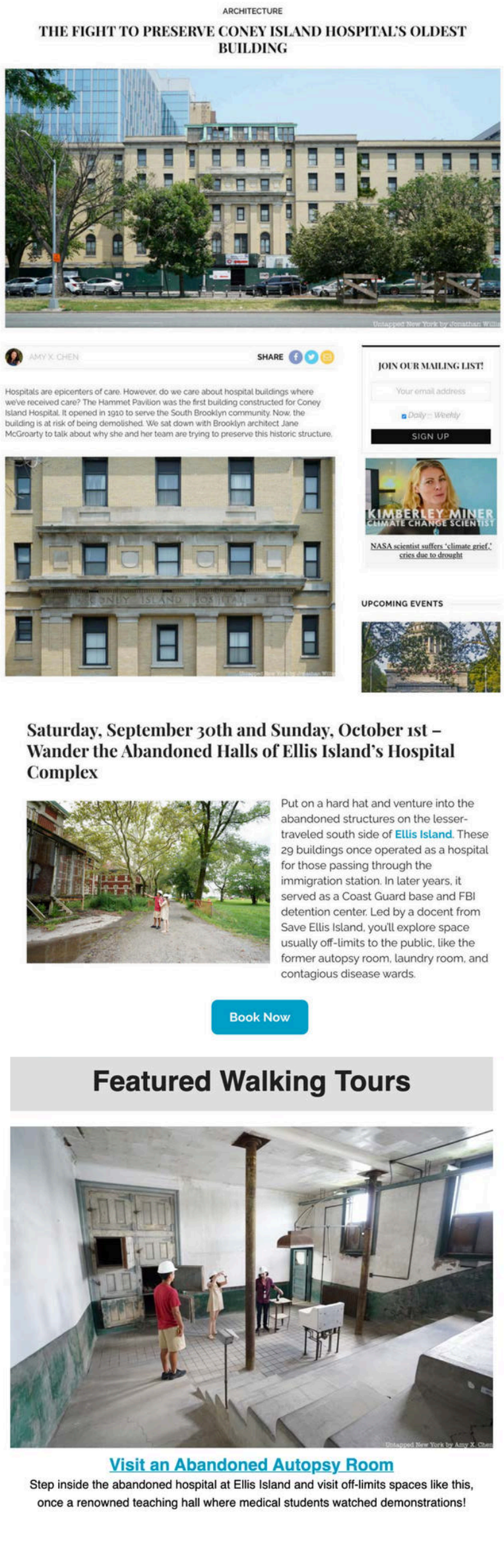
M3

URBAN JOURNALISM: UNTAPPED NEW YORK

During my internship for Untapped New York, I pitched and published articles and learned about New York City history through photographing Untapped Insider Tours including **Rockefellar Tour, Met Tour, Grand Central Tour, Brooklyn Subway Tour, Gilded Mansions Tour, and Ellis Island Hard Hat Tour**. I also photographed sites including **550 Madison, and the Battery Park Playscape**, and the **NYBG Summer Exhibit Opening**. When writing **"The Fight to Preserve Coney Island Hospital's Oldest Building,"** I interviewed Brooklyn architect Jane McGroarty about why their team is trying to preserve the Hammet Pavilion.

Project Type: Office-related
Editorial Journalism and Photography Internship
Untapped New York, New York
Date: May–June 2023

- "The Fight to Preserve Coney Island Hospital's Oldest Building"; "A Bold New Summer Art Exhibit Takes Over the New York Botanical Garden"; "10 Innovative Eco-Friendly Sites in Manhattan", Untapped New York, 2023



ARCHITECTURE

THE FIGHT TO PRESERVE CONEY ISLAND HOSPITAL'S OLDEST BUILDING

Untapped New York by Jonathan Willi

AMY X. CHEN SHARE

Hospitals are epicenters of care. However, do we care about hospital buildings where we've received care? The Hammet Pavilion was the first building constructed for Coney Island Hospital. It opened in 1910 to serve the South Brooklyn community. Now, the building is at risk of being demolished. We sat down with Brooklyn architect Jane McGroarty to talk about why she and her team are trying to preserve this historic structure.

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KIMBERLEY MINER CLIMATE CHANGE SCIENTIST

NASA scientist suffers 'climate grief,' cries due to drought

UPCOMING EVENTS



Untapped New York by Amy X. Chen



Untapped New York by Amy X. Chen

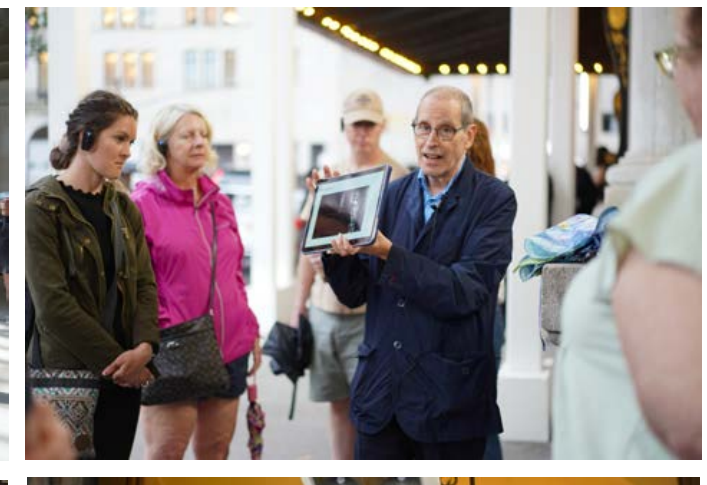
















Saturday, September 30th and Sunday, October 1st – Wander the Abandoned Halls of Ellis Island's Hospital Complex

Put on a hard hat and venture into the abandoned structures on the lesser-traveled south side of **Ellis Island**. These 29 buildings once operated as a hospital for those passing through the immigration station. In later years, it served as a Coast Guard base and FBI detention center. Led by a docent from Save Ellis Island, you'll explore space usually off-limits to the public, like the former autopsy room, laundry room, and contagious disease wards.

Book Now

Featured Walking Tours

Untapped New York by Amy X. Chen

Visit an Abandoned Autopsy Room

Step inside the abandoned hospital at Ellis Island and visit off-limits spaces like this, once a renowned teaching hall where medical students watched demonstrations!







New York Botanical Garden, Bronx

The New York Botanical Garden's major summer 2023 exhibit was "...things come to thrive...in the shedding...in the molting..." by Jamaican contemporary artist Ebony G. Patterson.

M4

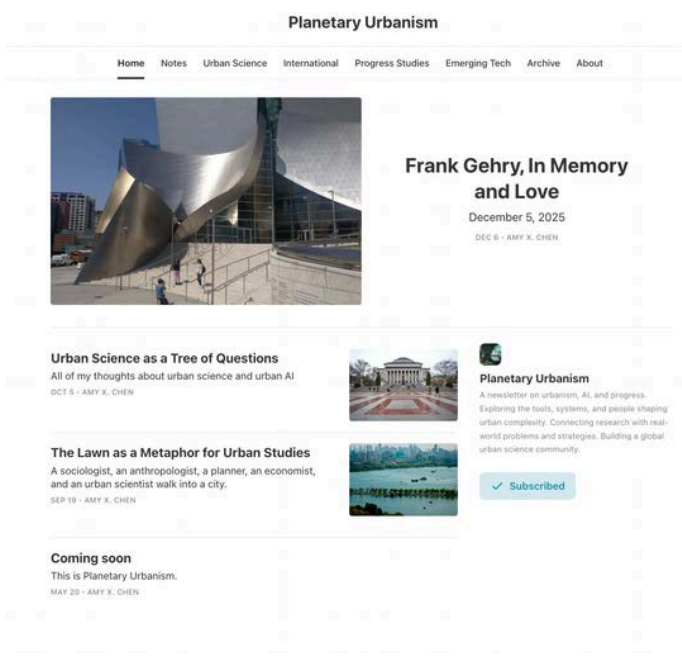
LOAD–BEARING & OTHER ESSAYS

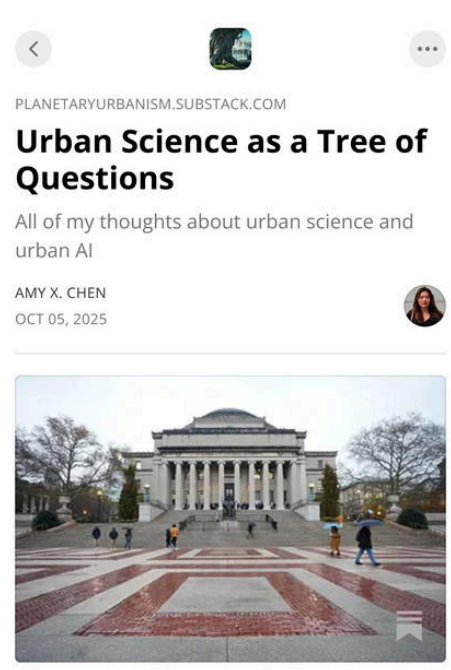
Project Type: Personal & Office-related
Substack; Columbia Daily Spectator; Urban AI; Barnard Columbia Urban Review; New Zealand Institute of Architects
Date: 2020 – 2025

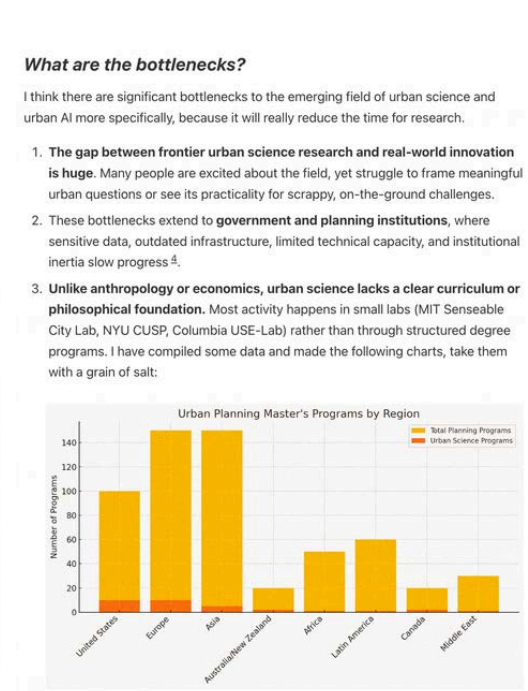
Writing always starts with a question and a vague but intriguing idea. To me, **writing is a critical process of inquiry**. Writing allows me to navigate through complex information and make my ideas coherent and compelling. Journalism has given me the skills to go out into the field, conduct interviews, and be rigorous about my research. Writing about an original idea takes time and good feedback loops; I want to become a better writer, which is why I started my personal substack a few months ago, and I plan to write more regularly in graduate school.

- Personal Substack: Load-Bearing
- “Governing Urban Artificial Intelligence: GovAI Coalition’s Vision for Responsible AI in the Public Sector”, Urban AI, 2024
- "Trucks, Fumes, and Groceries: A Tale of Environmental Injustice— Tracing Truck Exhaust in Mott Haven-Port Morris, South Bronx, New York", Barnard Columbia Urban Review Spring Edition, 2024
- "Peruvian Dream: A Fusion of Past, Present, and Future" in "10 Stories: Writing About Architecture", New Zealand Institute of Architects, 2020
- "Bringing Back Urban Swimming to Manhattan: A Deep Dive into the Plus Pool Project"; "With AI art, is the future already here?", Columbia Daily Spectator, 2022-2023

Personal Substack: Planetary Urbanism

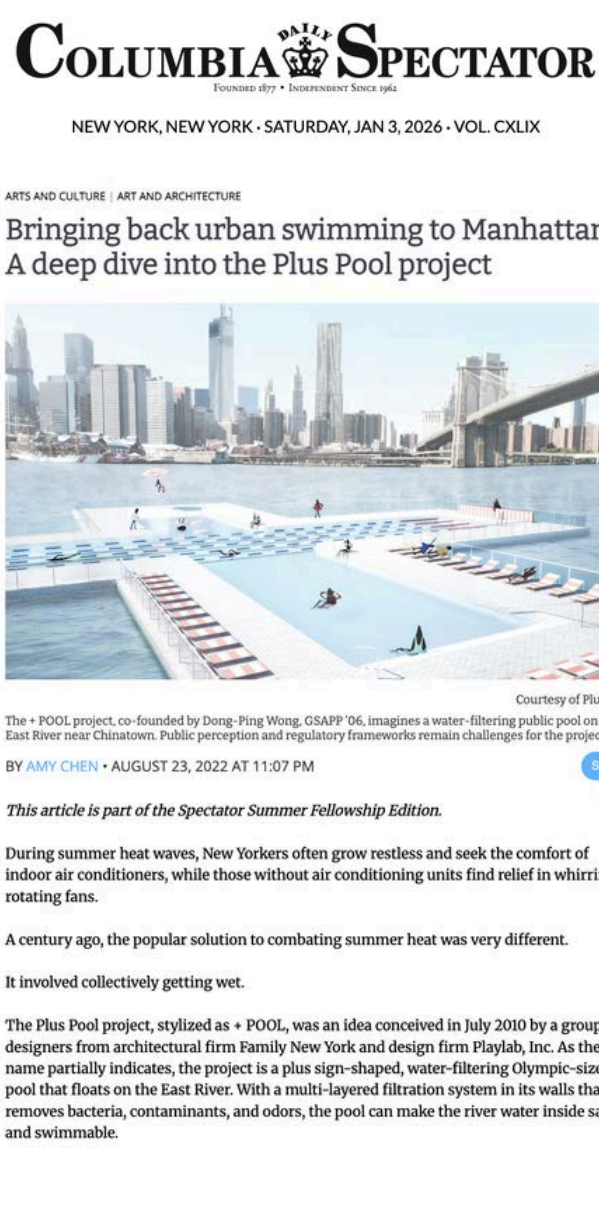


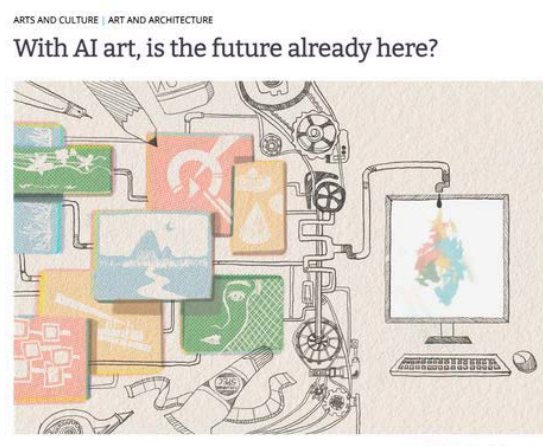


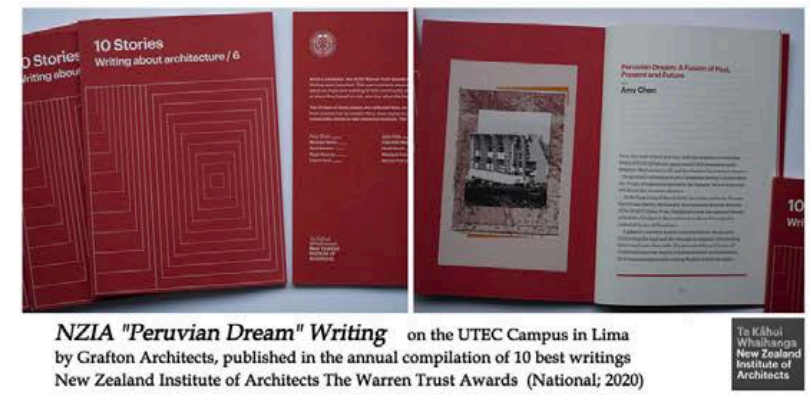




Publications and Journalism













THANK YOU

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